

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

L-3/T-1 CSE 300: Technical Writing and Presentation

Time: 45 minutes

Marks: 50

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1. Write down the most appropriate answer for each of the following multiple choice questions. (10x1=10)

Please format your answers in the following manner:

(a): (G)

(b): (K)

- (a) Which latex code will generate the correct output as follows:

this is a **bold** text with an *italic* text

(A) this is a `\bf{bold}` text with an `\textit{italic}` text

☒ (B) this is a `{\bfseries bold}` text with an `\emph{italic}` text

(C) this is a `\bfseries{bold}` text with an `\textit{italic}` text

(D) this is a `\bf{bold text}` with an `\em{italic}` text

- (b) Which documentclass should we use to write research papers?

☒ (A) article

(C) book

(B) report

(D) letter

- (c) Which is a good presentation practice?

☒ (A) Keep a small number of slides but assign enough time to each slide.

(B) Keep a large number of slides but assign enough time to only important slides.

(C) Keep a large number of slides and assign equal time to each slide.

(D) None of the above

- (d) Which of the following can not be a numbering scheme for reference items?

(A) Numbering in order of presence in the bibliography item list

(B) Numbering in order of mention

(C) Alphabet-wise

☒ (D) Random

- (e) Which of the following is not a list environment in LaTeX ?

(A) itemize

(C) enumerate

(B) description

☒ (D) record

- (f) Which of the following statements is true?

(A) `[htbp]` and `[pbth]` will produce an identical placement

(B) `[htbp]` and `[thbp]` will produce an identical placement

(C) Both (A) and (B)

☒ (D) None of the above

H → Here

T → Top

B → Bottom

P → Page

- (g) What are the minimum required components in a latex document?

☒ (A) `\documentclass{}`, `\begin{document}`, and `\end{document}`

(B) `\begin{document}` and `\end{document}`

(C) `\documentclass{}`, `\maketitle`, `\begin{document}`, and `\end{document}`

(D) None of the above

- (h) How do you write the column specifiers for a table with five columns, the first of which is left justified, the last of which is right justified, and the rest are centered? Except for the left and right boundaries, there are no vertical lines?

(A) `[lcccr]`

☒ (C) `{|l|c|c|c|r|}`

(B) `{|lcccr|}`

(D) `{lcccr}`

Needs | for boundary

! → avoids vertical line between them

- (i) What is the correct syntax for a LaTeX table?

☒ (A) `\begin{tabular}`

(C) `\begin{tables}`

(B) `\start{table}`

(D) All of the above

- (j) What will be the maximum allowed number of bibitems within the following bibliographic environment?

`\begin{thebibliography}{2}`

`\end{thebibliography}`

(A) 2

(C) 10

☒ (B) 9

(D) 99

Unlimited but the no. of digit should be always be multiple to 9.

for 3 digit → 999

for 2 digit → 99

2. Consider the given write-up `safelabeling.pdf` in which we have purposefully violated basic technical writing principles in many places. You have to identify and correct those. (20)

Please format your answers in the following manner.

- Line 2: “Dr.” is not allowed in authors’ names
- Line 48: lemma → Lemma
- etc.

3. Answer the following questions. (5x4=20)

(a) Suppose you want to generate a table in your CSE300 presentation (you can assume that the document class is beamer). Your table will have 3 rows and each row will have 3 columns. Your table (of size 3x3) will be blank on the first page (a particular frame) and will be populated with numbers having a maximum of 3 digits on the next page (the following frame). In order to keep the look consistent, although the table on the first page is blank, you want to keep the height and width of the cells identical to the populated table of the next page. For your convenience, the output is shown in Figure 2 (two consecutive pages i.e., frames). Note that the heights of the rows are also increased (2x of the default value). The code to generate the second frame with the populated table is also given. Now write necessary codes in order to generate the output of the first frame having the blank table. In case you find it difficult to write the exact code, try to write an idea to accomplish this task. This could be worth some partial credit.

To increase, use `\arraystretch{4}`
For increasing width, `\phantom{1000}`

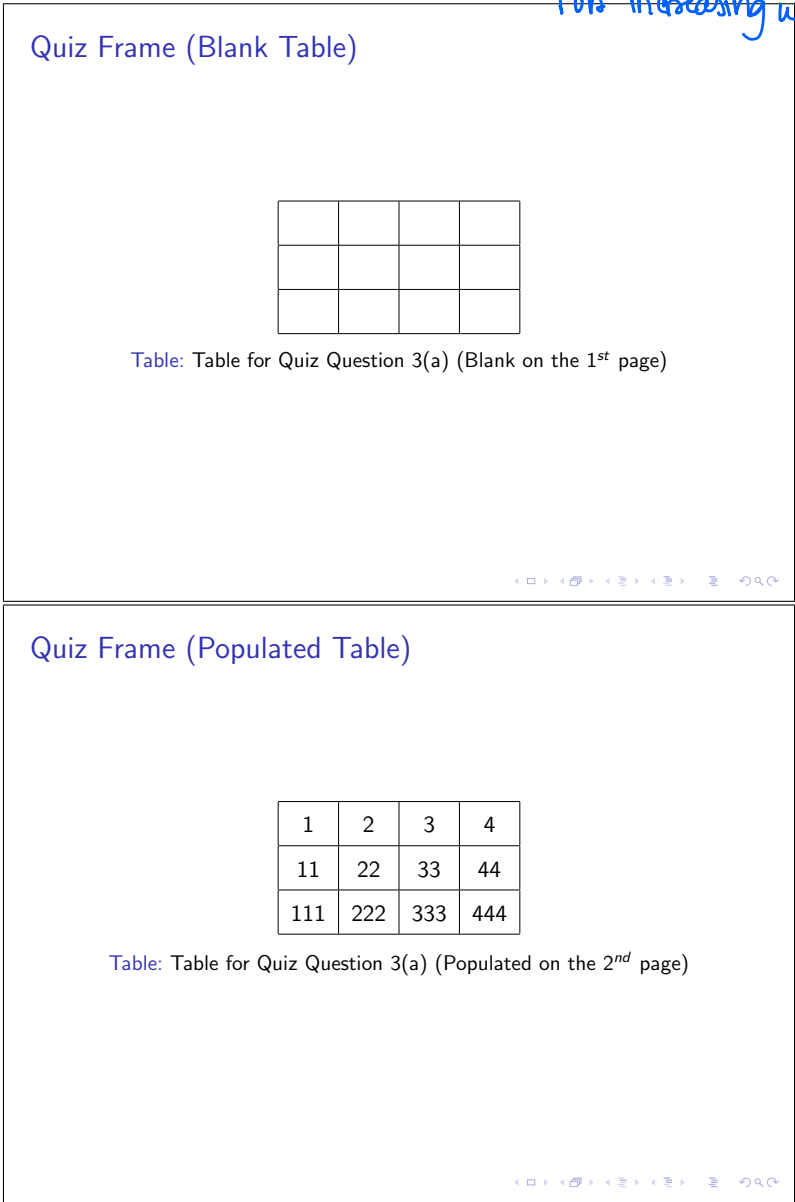


Figure 1: Figure of two consecutive frames for Question No. 3(a)

```

1 \begin{frame}{Quiz Frame (Populated Table)}
2   \begin{table}[]
3     \centering
4     \begin{tabular}{|c|c|c|c|}
5       \hline
6       1 & 2 & 3 & 4 \\
7       \hline
8       11 & 22 & 33 & 44 \\
9       \hline
10      111 & 222 & 333 & 444 \\
11      \hline
12    \end{tabular}
13    \caption{Table for Quiz Question 3(a) (Populated on the $2^{\text{nd}}$ page)}
14    \label{tab:my_label}
15  \end{table}
16 \end{frame}

```

Figure 2: Figure for code to generate the second frame of Question 3(a)

- (b) Why should we use $\text{\LaTeX}$ instead of WYSIWYG (What You See Is What You Get) editors e.g., MS Word, LibreOffice Writer, Apple Pages? State three concrete reasons with justifications.
 - ① Cross reference
 - ② Scientific type
 - ③ Various packages
- (c) When do we need to use vector graphics instead of raster graphics in technical writing/presentation? Give one example for each when you need to use pie-chart rather than bar chart and vice-versa.
 - vector → one page, multiple content
 - one page per frame → pie-chart
 - discrete quantities → bar chart
- (d) In beamer, how do slides and frames differ? Give an example. Mention two points which help in making your presentation slides visible from a distance.
 - ① Font colors & size
 - ② Don't do thoroughly writing, adding list & items
- (e) What will be outcome of the following LaTeX code snippet?

Consider the following equation.

```

\begin{equation}
a + b = c
\end{equation}

```

It can also be written as follows: $\$a+b=c\$$ If you prefer, it can also be written as $\$a+b=c\$$.

Consider the following equation.

$$a+b=c$$

It can also be written as follows:

$$a+b=c$$

If you prefer, it can also be written as $a+b=c$

Safe Labeling of Graphs with Minimum Span

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Abstract. Let $G = (V, E)$ be a graph with vertex set V and edge set E such that $|V| = n$ and $|E| = m$. $G' = (V', E')$ is a subgraph of G if $V = V'$, $E' \subseteq E$. I wish to label all vertices of G with a positive integer such that each vertex will receive a distinct integer and the difference of the labels of two adjacent vertices in G will be at least k . We will call such a labeling of G a k -safe labeling of G . The range from the smallest to the largest integers assigned to the vertices of G in a k -safe labeling is called the span of the k -safe labeling. k -safe labeling problem finds a k -safe labeling of a graph with the minimum span. In this paper I will give upper bounds on k -safe labelings of trees and bipartite graphs. k -safe labeling has many practical applications in radio frequency assignment [4, 5].

Keywords: Safe labeling, Minimum, Small Span, Bipartite Graphs.

1 Introduction

Graph labeling is an interesting research topic in graph theory which concerns the assignment of values to the edges and/or vertices of a graph such that the assignments meet some conditions in the graph[1]. Let G be a graph of n vertices and k be a positive integer. I wish to label the vertices of G with positive integers such that each vertex receives a distinct integer and the difference of the labels of two adjacent vertices is at least k . Call such a labeling of G a k -safe labeling of G . We call the range from the largest to the smallest integers assigned to the vertices of G in a k -safe labeling the *span* of the k -safe labeling. $I_l - I_s + 1$ is the span of the k -safe labeling of G , where I_s and I_l be the smallest and the largest integer.

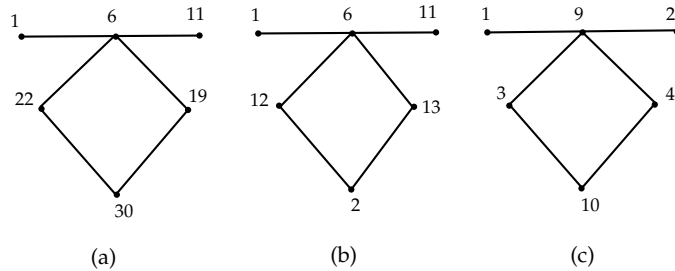


Fig. 1. For $k = 5$ (a) labeling of a graph with span 30, (b) labeling of a graph with span 13 and (c) labeling of a graph with minimum span 10.

25 The k -safe labeling problem asks to find a k -safe labeling of a graph with the minimum span .
 26 In figure 1 we show three k -safe labelings of the same graph of 6 vertices. The spans of the k -safe
 27 labelings in figure 1(a), (b) and (c) are thirty, 13 and 10, respectively . In fact the k -safe labeling
 28 in Figure 1(c) is a k -safe labeling with the minimum span. K_n needs span at least $(n-1)k+1$ ^{reference?}
 29 for a k -safe labeling. Again for a complete bipartite graph $K_{p,q}$ the span for a k -safe labeling is at
 30 least $p+q+k-1 = n+k-1$, since $p+q = n$. Thus one may expect to find k -safe labelings of
 31 sparse graphs using narrow spans.

32 2 Preliminaries

33 Let $G = (V, E)$ be a simple graph . A *path* is a sequence $v_1, v_2, \dots, v_q$ of distinct vertices (except
 34 possibly v_1 and v_q) such that $v_1, \dots, v_q \in V$ and $(v_{i-1}, v_i) \in E$ for all $2 \leq i \leq q$. The *length* of the
 35 path is the number of edges on the path. A path $v_1, v_2, \dots, v_q$ is *closed* if $v_1 = v_q$. A closed path
 36 containing at least one edge is called a *cycle*. An *odd cycle* is a cycle with odd length, that is, with
 37 an odd number of edges . An *even cycle* is a cycle with even length, that is, with an even number
 38 of edges. A graph G is *connected* if there is a path between every pair of vertices in G . If G is a
 39 connected graph containing no cycle then G is called a tree T .

40 Let $G = (V, E)$ be a graph. A subset V' of V is called *independent* if the vertices in V' are
 41 pair-wise non-adjacent. If the vertex set V of G can be partitioned into two nonempty subsets
 42 P and Q such that both P and Q are independent sets then G is called a *bipartite graph*. The
 43 partition $V = P \cup Q$ is called a *bipartition* of the bipartite graph G . Note that $P \cup Q = V$ and
 44 $P \cap Q = \emptyset$. The following Lemmas on bipartite graphs are known [6].

45 **Lemma 1.** G is bipartite if and only if G has no odd cycle.

46 **Lemma 2.** A graph G is bipartite if and only if the vertices of G ^{G} can be colored by two colors such
 47 that every pair of adjacent vertices receive different colors.

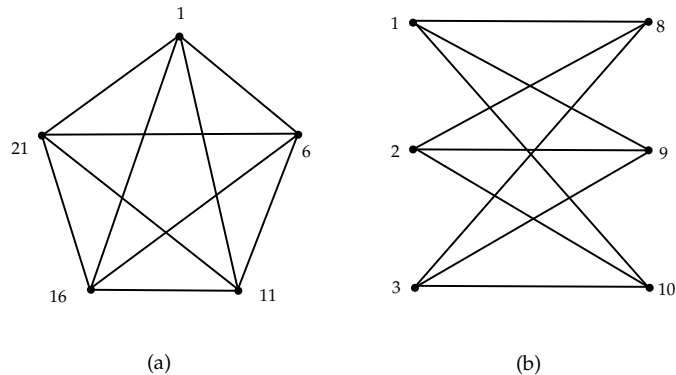


Fig. 2. For $k = 5$ (a) A k -safe labeling of K_5 with span 21, (b) a k -safe labeling of $K_{3,3}$ with span 10.

48 The characterization of bipartite graphs in lemma 2, leads to a linear-time algorithm to deter-
 49 mine whether a graph is bipartite or not, and also compute a bipartition of a bipartite graph.

50 3 k -safe Labelings of Sparse Graphs

51 **Theorem 1.** *Let $G = (V, E)$ be a bipartite graph of n vertices with the bipartition $V = P \cup Q$.
 52 Then G admits a k -safe labeling with a span $n + k - 1$.*

53 Weak It is interesting to observe that the trivial span $n + k - 1$ in theorem 1 is optimal for bipartite
 54 graphs, since a complete bipartite graph does not admit a k -safe labeling with a span smaller than
 55 $n + k - 1$.

56 We have the following theorem on k -safe labelings of trees.

57 **Theorem 2.** *Let T be a tree of n vertices. Let m be the number of edges in T and $\Delta(T)$ is the
 58 maximum degree of T . Every tree of n vertices admits a k -safe labeling with span $n + k - 1$.*

Proof. Since a tree has no cycle, a tree is a bipartite graph by lemma 1 in section 2. Hence every
 tree of n vertices has a k -safe labeling with span $n + k - 1$ by theorem 1. $\square$

59 4 Conclusions

60 We study safe labeling here which has many applications. Noice flex bras! So this is a great paper.

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The Complete CSE 300 Study Guide

LaTeX • TikZ • Beamer • Technical Writing • Research Papers

From beginner to advanced, with every example compiled

Built from your CSE 300 slides, quiz and lectures

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Part I

LaTeX from Zero to Advanced

Chapter 1

How LaTeX Works

1.1 What LaTeX is, and why we use it

LaTeX is a *markup* system. You write plain text with commands that describe *what* each piece is (a section title, an equation, a citation) and LaTeX decides *how* it looks. A WYSIWYG editor (Word, LibreOffice, Pages) works the other way: you format by hand as you type.

↳ what you see is what you get

Quiz question: why LaTeX over WYSIWYG? (3 reasons with justification)

1. **Superior mathematical typesetting.** Equations, symbols and alignment are typeset to professional (publisher) quality; Word equations are harder to write, align and keep consistent.
2. **Automatic numbering and cross-referencing.** Sections, figures, tables, equations and citations are numbered by `\label/\ref/\cite`. Insert a figure and every number updates; in Word you fix them by hand.
3. **Consistent, professional formatting separated from content.** The document class and conference templates (IEEE, ACM, Springer LNCS) control layout, so you focus on content and can switch formats by changing one line.

Other valid reasons: plain-text files work with version control (Git) and collaboration (Overleaf); free and cross-platform; BibTeX handles reference styles automatically; stable output for large documents such as theses.

1.2 The compile cycle

A `.tex` file is compiled into a PDF by an engine:

- pdflatex: the standard engine; fast, fine for English.
- xelatex / lualatex: support Unicode and system fonts (needed for Bangla text, e.g. with `fontspec`).

LaTeX reads the file top to bottom once. Anything that needs information from *later* in the document (a reference to Figure 5, the table of contents, citations) is written to auxiliary files (`.aux`, `.toc`, `.bbl`) and used on the *next* run. That is why you sometimes see “??” and must compile twice. With BibTeX the sequence is pdflatex → bibtex → pdflatex → pdflatex. Overleaf and latexmk do this automatically.

1.3 The minimum document

```
\documentclass{article} % preamble starts: global settings
\begin{document} % body starts: the actual content
Hello, world!
\end{document} % anything after this is ignored
```

Exam focus

The *minimum required components* are exactly `\documentclass{}`, `\begin{document}` and `\end{document}`. `\maketitle` is optional. (Quiz 1(g), answer A.)

Everything between `\documentclass` and `\begin{document}` is the **preamble**: you load packages and define commands there. Nothing in the preamble is printed.

1.4 Anatomy of commands

- **Commands** start with a backslash: `\today`, `\LaTeX`. Names are letters only and case-sensitive.
- **Mandatory arguments** go in braces: `\textbf{bold}`.
- **Optional arguments** go in square brackets, before the mandatory ones: `\documentclass[12pt]{article}`.
- **Environments** have a begin and an end: `\begin{itemize} ... \end{itemize}`. Everything inside is affected.
- **Groups**: braces { ... } limit the effect of *declarations* such as `\bfseries` or `\large`.
- **Comments** start with % and run to the end of the line.
- **Starred versions**: `\section*{}` (unnumbered), `\l* (no page break)`. The star changes behaviour.

A command name ends at the first non-letter, and spaces after it are swallowed: `\LaTeX is` prints “ $\text{\LaTeX}$ is”. Write `\LaTeX{}` is or `\LaTeX\` is to keep the space.

1.5 Document classes

Class	Use	Has <code>\chapter</code> ?
<code>article</code>	research papers, short reports, assignments	No
<code>report</code>	longer reports, theses, lab reports	Yes
<code>book</code>	books (two-sided, front/back matter)	Yes
<code>letter</code>	letters	No
<code>beamer</code>	presentation slides	No
<code>IEEEtran</code> , <code>llncs</code> , <code>acmart</code>	conference/journal templates	No

Exam focus

“Which documentclass should we use to write research papers?” → `article` (Quiz 1(b)).

Common class options: `10pt/11pt/12pt` (base font size), `a4paper/letterpaper`, `twocolumn`, `twoside`, `landscape`, `draft` (marks overfull lines, skips images), `titlepage`.

1.6 Essential packages

Package	What it gives you
amsmath, amssymb, amsthm, mathtools	advanced maths, symbols, theorem environments
graphicx	<code>\includegraphics</code> for images
geometry	page margins: <code>\usepackage [margin=2cm]{geometry}</code>
hyperref	clickable links, <code>\href</code> , <code>\url</code> (load near the end)
cleveref	<code>\cref</code> writes “Figure 3” automatically (load after hyperref)
booktabs	professional table rules <code>\toprule</code> , <code>\midrule</code> , <code>\bottomrule</code>
* multirow, tabularx, array, longtable	advanced tables
xcolor	colours: <code>\textcolor</code> , <code>\colorbox</code>
* enumitem	customise lists
* subcaption	sub-figures (a), (b)
tikz, pgfplots	drawing and plotting
* listings, minted	source code
* algorithm2e, algorithmicx	pseudocode
siunitx	numbers and units: <code>\SI {3.5}{\giga \hertz }</code>
natbib, biblatex	bibliography management
float	the [H] placement specifier
fontspec	system fonts (XeLaTeX/LuaLaTeX only)

1.7 Reading error messages

- Undefined control sequence: typo in a command, or a missing package.
- Missing \$ inserted: a maths symbol (`_`, `^`, `\alpha`) used outside maths mode.
- Missing } inserted / Runaway argument: unbalanced braces.
- Environment ... undefined: typo or missing package.
- File not found: wrong image or file path.
- Overfull \hbox: a line sticks into the margin (a warning, not an error).
- Citation/Reference undefined: compile again, or check the key spelling.

Tip

Read the *first* error only; later errors are usually caused by it. The line number 1.42 tells you where LaTeX noticed it.

Chapter 2

Text, Fonts and Spacing

2.1 Paragraphs and line breaks

A blank line starts a new paragraph. Several spaces count as one space; one newline counts as a space.

```
First paragraph. Extra spaces
are ignored.

Second paragraph.\\
Forced line break with \verb|\\|.
\par Another paragraph via \verb|\par|.
```

First paragraph. Extra spaces are ignored.
Second paragraph.
Forced line break with \\
Another paragraph via \par.

Common mistake

Do not use \\ to separate paragraphs or add space; use a blank line and \vspace. \newpage starts a new page; \clearpage also flushes waiting figures.

2.2 Special characters

Ten characters are reserved: # \$ % ^ & _ { } ~ \.

```
\# \$ \% \& \_ \{ \} ~ \\
\textasciitilde{} \textasciicircum{} \textbackslash \\
Quotes: ``double'' and `single' \\
Dashes: - (hyphen), -- (range 1--5), --- (em dash) \\
Dots: \ldots\ and non-breaking: Figure~1
```

\$ % & _ { }
~ ^ \
Quotes: “double” and ‘single’

Dashes: - (hyphen), – (range 1–5), — (em dash)
Dots: ... and non-breaking: Figure 1

Use ‘ ‘ and ’ ’ for quotation marks, never ". The tilde ~ is a non-breaking space: write Figure~\ref{f} and Dr.~Rahman so the number never starts a new line.

2.3 Font styles: commands vs declarations

There are two kinds of font commands, and the quiz tests this directly.

- **Commands with an argument** affect only their argument: `\textbf{...}`, `\textit{...}`, `\emph{...}`.
- **Declarations** affect everything after them until the group ends: `{\bfseries ...}`, `{\itshape ...}`, `{\em ...}`.

```
\textbf{bold} \textit{italic} \textsl{slanted}
\textsc{Small Caps} \texttt{typewriter}
\textsf{sans serif} \textrm{roman}
\underline{underlined} \emph{emphasis}
```

```
{\bfseries bold group} normal again \\
{\itshape italic group} normal \\
\textbf{\textit{bold italic}} \\
\emph{Emph \emph{nested} toggles}
```

bold italic slanted SMALL CAPS typewriter sans serif roman underlined *emphasis*
bold group normal again
italic group normal
bold italic
Emph nested toggles

Effect	Command (argument)	Declaration (group)
Bold	<code>\textbf{...}</code>	<code>{\bfseries ...}</code>
Italic	<code>\textit{...}</code>	<code>{\itshape ...}</code>
Emphasis	<code>\emph{...}</code>	<code>{\em ...}</code>
Typewriter	<code>\texttt{...}</code>	<code>{\ttfamily ...}</code>
Sans serif	<code>\textsf{...}</code>	<code>{\sffamily ...}</code>
Small caps	<code>\textsc{...}</code>	<code>{\scshape ...}</code>
Upright	<code>\textup{...}</code>	<code>{\upshape ...}</code>

Quiz 1(a): “this is a bold text with an *italic* text”

- (A) `\bf{bold}`: `\bf` is an obsolete *declaration*; braces after it do not limit it, so everything after becomes bold.
- (B) `{\bfseries bold} ... \emph{italic}`: **correct**. The group limits `\bfseries`, and `\emph` italicises its argument.
- (C) `\bfseries{bold}`: same problem as A; `\bfseries` is a declaration, not a command with an argument.

- (D) `\em{italic}`: `\em` is a declaration; everything after it is emphasised.

Rule: a declaration must be *inside* the braces, `{\bfseries x}`, not `\bfseries{x}`.

`\emph` is preferred over `\textit` for emphasis because it is *semantic*: inside italic text it switches back to upright.

2.4 Font sizes

```
{\tiny tiny} {\scriptsize scriptsize}
{\footnotesize footnotesize} {\small small}
{\normalsize normalsize} {\large large}
{\Large Large} {\LARGE LARGE}
{\huge huge} {\Huge Huge}
```

tiny scriptsize footnotesize small normalsize large Large LARGE huge Huge

Size commands are declarations, so always use them inside a group or an environment.

2.5 Colour

```
\textcolor{red}{Red text},
\textcolor{blue!60!black}{dark blue},
\textcolor{green!50}{50\% green}. \\
\colorbox{yellow}{highlighted} \\
\fcolorbox{red}{white}{framed box}
```

↳ border ↳ inside

Red text, dark blue, 50% green.

highlighted

framed box

`blue!60!black` means 60% blue mixed with 40% black. Define your own with `\definecolor{buet}{RGB}{0,82,`

2.6 Horizontal and vertical space

```
A\hspace{1cm}B \\
A\quad B\qquad C \\
A\,B\,C\ D (thin, medium, normal) \\
left\hfill right \\
A\hphantom{XXX}B (invisible XXX)
\vspace{3mm}
```

After vertical space.


```
\end{description}
```

- Bullet
 - Nested
- 1. First
- 2. Second

CPU Processor

RAM Memory

Exam focus

The three list environments are `itemize`, `enumerate` and `description`. `record` is *not* a list environment (Quiz 1(e)).

With `enumitem` you control labels and spacing:

```
\begin{enumerate}[label=(\alph*)]
  \item alpha
  \item beta
\end{enumerate}
\begin{enumerate}[label=\Roman*.,itemsep=0pt]
  \item Roman
  \item numerals
\end{enumerate}
\begin{itemize}[label=$\star$]
  \item custom bullet
\end{itemize}
```

(a) alpha

(b) beta

I. Roman

II. numerals

★ custom bullet

Label counters: `\arabic*` (1), `\alph*` (a), `\Alph*` (A), `\roman*` (i), `\Roman*` (I). Use `[resume]` to continue numbering after an interruption.

2.9 Footnotes, quotes and verbatim

Text with a note. `\footnote`{The footnote.}

```
\begin{quote}
A short indented quotation.
```

```
\end{quote}
```

```
\verb|\textbf{raw}| shown literally.
```

```
\begin{verbatim}
```

```
for i in range(3):
```

```
    print(i)
```

```
\end{verbatim}
```

Text with a note.^a

A short indented quotation.

```
\textbf{raw} shown literally.
```

```
for i in range(3):
```

```
    print(i)
```

^aThe footnote.

`quote` is for short quotations, `quotation` for multi-paragraph ones. `\verb` uses any delimiter pair, for example `\verb+x_1+`; it cannot be used inside the argument of another command.

2.10 Links

```
\url{https://www.buet.ac.bd} \\\
```

```
\href{https://overleaf.com}{Overleaf} \\\
```

```
\href{mailto:a@b.com}{email me}
```

<https://www.buet.ac.bd>

Overleaf

email me

Chapter 3

Document Structure

3.1 Title, author, abstract

```
\documentclass{article}
\title{Safe Labeling of Graphs with Minimum Span}
\author{Umme Habiba\thanks{Department of CSE, BUET, Dhaka-1205}
\and Second Author}
\date{\today} % or \date{} for no date
\begin{document}
\maketitle % prints title, author, date
\begin{abstract}
One paragraph summarising problem, method, result.
\end{abstract}
\end{document}
```

`\thanks` puts affiliation or funding in a footnote; `\and` separates authors. Never write “Dr.” or “Prof.” in author names in a paper.

3.2 Sectioning

Command	Level	Available in
<code>\part{}</code>	−1	all
<code>\chapter{}</code>	0	report, book
<code>\section{}</code>	1	all
<code>\subsection{}</code>	2	all
<code>\subsubsection{}</code>	3	all
<code>\paragraph{}</code>	4	all
<code>\subparagraph{}</code>	5	all

- `\section*{Acknowledgement}` is unnumbered and not in the table of contents.
- `\section[Short]{A Very Long Title}`: “Short” appears in the table of contents.
- `\tableofcontents`, `\listoffigures`, `\listoftables` generate lists (compile twice).
- `\appendix` switches numbering to A, B, C.
- In a book: `\frontmatter` (roman page numbers), `\mainmatter`, `\backmatter`.

- `\setcounter{secnumdepth}{2}` and `\setcounter{tocdepth}{1}` control numbering and TOC depth.

3.3 Labels and cross-references

Demo text.
`\begin{equation}`
 $E = mc^2$ `\label{eq:einstein}`
`\end{equation}`
Equation~`\eqref{eq:einstein}` and
Equation~`\ref{eq:einstein}`;
with cleveref: `\cref{eq:einstein}`.
This is on page~`\pageref{eq:einstein}`.

Demo text.

$$E = mc^2 \tag{3.1}$$

Equation (3.1) and Equation 3.1; with cleveref: eq. (3.1). This is on page 17.

- `\label{key}` remembers the *last numbered thing*. In a figure, put `\label` **after** `\caption`, or you get the section number.
- `\ref` gives the number, `\eqref` adds parentheses, `\pageref` gives the page, `\cref` adds the word (“Figure”, “eq.”).
- Use prefixes: `fig:`, `tab:`, `eq:`, `sec:`, `thm:`, `lem:`.

3.4 Splitting a big project

```
\input{chapters/intro} % pastes the file here (can nest)
\include{chapters/method} % new page; works with \includeonly
\includeonly{chapters/method} % in preamble: compile only this
```

A thesis typically has `main.tex`, a `chapters/` folder, a `figures/` folder and `refs.bib`.

3.5 Page layout, headers and page numbers

```
\usepackage[a4paper, margin=2.5cm]{geometry}
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{} % clear defaults
\fancyhead[L]{CSE 300}
\fancyhead[R]{\leftmark} % current section
\fancyfoot[C]{\thepage}
\pagenumbering{roman} % i, ii, iii (then arabic)
\linespread{1.3} % line spacing
\setlength{\parindent}{0pt} % no paragraph indent
\setlength{\parskip}{6pt} % space between paragraphs
```

```
\thispagestyle{empty} % no number on this page
```

Page styles: `plain` (number only), `empty`, `headings`, `fancy`. Use `\begin{multicols}{2}` (package `multicol`) for columns within a page, or the `twocolumn` class option.

Chapter 4

Mathematics

4.1 Inline and display maths

Inline: $a^2+b^2=c^2$ or (x_i) .

Display, unnumbered:

$\left[\sum_{i=1}^n i = \frac{n(n+1)}{2} \right]$

Numbered:

$$a+b=c \quad \text{\label{eq:abc}}$$

Unnumbered with star:

$$x = 1$$

Inline: $a^2 + b^2 = c^2$ or x_i .

Display, unnumbered:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

Numbered:

$$a + b = c \tag{4.1}$$

Unnumbered with star:

$$x = 1$$

Avoid $\$a+b=c\$$

$\$a+b=c\$$ is plain-Tex syntax. It still works, but gives wrong spacing and ignores class options such as `fleqn`. Use $\left[\dots \right]$ instead.

Quiz 3(e): the output of the snippet

Consider the following equation.

```
\begin{equation}
```

$$a + b = c$$

```
\end{equation}
```

It can also be written as follows: $a+b=c$ If you prefer,

it can also be written as $a+b=c$.

Output: (1) “Consider the following equation.” then $a + b = c$ centred on its own line with the number (1) at the right; (2) “It can also be written as follows:” then $a + b = c$ centred again but **unnumbered**; (3) “If you prefer, it can also be written as $a + b = c$.” inline in the sentence. The spaces inside maths are ignored, so $a + b = c$ and $a+b=c$ look identical.

4.2 Superscripts, subscripts, fractions, roots

```
 $x^2$ ,  $x^{10}$ ,  $x_i$ ,  $x_{i,j}$ ,  
 $x_i^2$ ,  $e^{-x^2}$  \\  
 $\frac{a}{b}$ ,  $\dfrac{a}{b}$ ,  $\tfrac{a}{b}$  \\  
 $\sqrt{x}$ ,  $\sqrt[3]{x}$  \\  
 $\binom{n}{k}$ ,  $\overline{AB}$ ,  $\hat{x}$ ,  
 $\vec{v}$ ,  $\bar{x}$ ,  $\dot{x}$ ,  $\tilde{y}$ 
```

$x^2, x^{10}, x_i, x_{i,j}, x_i^2, e^{-x^2}$
 $\frac{a}{b}, \frac{a}{b}, \frac{a}{b}$
 $\sqrt{x}, \sqrt[3]{x}$
 $\binom{n}{k}, \overline{AB}, \hat{x}, \vec{v}, \bar{x}, \dot{x}, \tilde{y}$

Braces group more than one character: x^10 gives x^{10} (wrong), $x^{\{10\}}$ gives x^{10} .

4.3 Greek letters and symbols

α	<code>\alpha</code>	β	<code>\beta</code>	γ, Γ	<code>\gamma, \Gamma</code>
δ, Δ	<code>\delta, \Delta</code>	ϵ, ε	<code>\epsilon, \varepsilon</code>	θ	<code>\theta</code>
λ	<code>\lambda</code>	μ	<code>\mu</code>	π	<code>\pi</code>
σ, Σ	<code>\sigma, \Sigma</code>	ϕ, φ	<code>\phi, \varphi</code>	ω, Ω	<code>\omega, \Omega</code>
$\leq$	<code>\le</code>	$\geq$	<code>\ge</code>	$\neq$	<code>\ne</code>
$\approx$	<code>\approx</code>	$\equiv$	<code>\equiv</code>	$\sim$	<code>\sim</code>
$\in$	<code>\in</code>	$\notin$	<code>\notin</code>	$\subseteq$	<code>\subseteq</code>
$\cup$	<code>\cup</code>	$\cap$	<code>\cap</code>	$\emptyset$	<code>\emptyset</code>
$\forall$	<code>\forall</code>	$\exists$	<code>\exists</code>	$\neg$	<code>\neg</code>
$\wedge$	<code>\wedge</code>	$\vee$	<code>\vee</code>	∞	<code>\infty</code>
$\rightarrow$	<code>\to</code>	$\Rightarrow$	<code>\Rightarrow</code>	$\Leftrightarrow$	<code>\iff</code>
$\times$	<code>\times</code>	$\cdot$	<code>\cdot</code>	$\pm$	<code>\pm</code>
$\dots$	<code>\ldots</code>	$\cdots$	<code>\cdots</code>	$\vdots, \ddots$	<code>\vdots, \ddots</code>
$\lfloor x \rfloor$	<code>\lfloor x \rfloor</code>	$\lceil x \rceil$	<code>\lceil x \rceil</code>	∂	<code>\partial</code>
$\mathbb{R}$	<code>\mathbb{R}</code>	$\mathcal{O}$	<code>\mathcal{O}</code>	∇	<code>\nabla</code>

`\ldots` dots sit on the baseline (lists: $1, \dots, n$); `\cdots` are centred (operations: $1 + \dots + n$).

4.4 Big operators, functions and delimiters

```
\[ \sum_{i=1}^n a_i \quad \prod_{k=1}^n k
\quad \int_0^1 f(x) \, dx \quad \lim_{x \rightarrow 0} \ ]
\[ \sin x, \log n, \max_{v \in V} d(v), \
O(n \log n) \ ]
\[ \left( \frac{a}{b} \right)
\left[ \sum x \right]
\left\{ x \mid x > 0 \right\}
\left| \frac{1}{2} \right| \ ]
```

$$\sum_{i=1}^n a_i \quad \prod_{k=1}^n k \quad \int_0^1 f(x) dx \quad \lim_{x \rightarrow 0}$$

$$\sin x, \log n, \max_{v \in V} d(v), O(n \log n)$$

$$\left(\frac{a}{b}\right) \left[\sum x\right] \{x \mid x > 0\} \left|\frac{1}{2}\right|$$

- Write `\sin`, `\log`, `\max`, not `sin`: plain letters are italic variables (*sin* vs `sin`).
- `\left` and `\right` scale delimiters to their content; use `\left.` for an invisible side.
- Define new operators: `\DeclareMathOperator{\argmax}{arg\,max}` in the preamble.
- `\,` before `dx` adds a thin space.

4.5 Text and fonts inside maths

```
$x = 1 \text{ if } y > 0$ \
$\mathbf{v}$, $\mathrm{d}x$, $\mathbb{N}$,
$\mathcal{L}$, $\mathit{diff}$ \
$\boldsymbol{\alpha}$ (needs amsmath)
```

$x = 1$ if $y > 0$
 $\mathbf{v}$, dx , $\mathbb{N}$, $\mathcal{L}$, $diff$
 $\boldsymbol{\alpha}$ (needs amsmath)

4.6 Multi-line equations

```
\begin{align}
f(x) &= (x+1)^2 \ \\
&= x^2 + 2x + 1 \nonumber \ \\
&\geq 0
\end{align}
```

```
\begin{align*}
a &= 1 & b &= 2
\end{align*}
```

$$f(x) = (x + 1)^2 \tag{4.2}$$

$$= x^2 + 2x + 1$$

$$\geq 0 \tag{4.3}$$

$$a = 1$$

$$b = 2$$

- `align`: `&` marks the alignment point, `\\` ends a line. Every line is numbered; `\nonumber` or `\notag` suppresses one; `align*` numbers none.
- `gather`: several centred equations, no alignment.
- `multline`: one long equation broken over lines.
- `split` inside `equation`: one number for several aligned lines.
- Never use the old `eqnarray`; its spacing is wrong.

```
\begin{equation}
\begin{split}
T(n) &= 2T(n/2) + n \\
&= O(n \log n)
\end{split}
\end{equation}
\[ |x| = \begin{cases}
x & \text{if } x \geq 0 \\
-x & \text{otherwise}
\end{cases}
\]
```

$$T(n) = 2T(n/2) + n$$

$$= O(n \log n) \tag{4.4}$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{otherwise} \end{cases}$$

4.7 Matrices

```
\[ \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}
\begin{bmatrix} a & b \\ c & d \end{bmatrix}
\begin{vmatrix} a & b \\ c & d \end{vmatrix} \]
\[ A = \begin{pmatrix}
a_{11} & \cdots & a_{1n} \\
\vdots & \ddots & \vdots
\end{pmatrix}
```

```
a_{m1} & \cdots & a_{mn} \\
\end{pmatrix} \]
```

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \left| \begin{array}{c} a & b \\ c & d \end{array} \right|$$

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

`matrix` (no brackets), `pmatrix` (), `bmatrix` [], `Bmatrix` { }, `vmatrix` | , `Vmatrix` || ||.

4.8 Theorems, lemmas and proofs

Define theorem-like environments in the preamble with `\newtheorem`:

```
\usepackage{amsthm}
\newtheorem{theorem}{Theorem}[section] % numbered 2.1, 2.2 within sections
\newtheorem{lemma}[theorem]{Lemma} % shares the theorem counter
\newtheorem{corollary}{Corollary}[theorem]
\theoremstyle{definition} % upright body text
\newtheorem{definition}{Definition}
\theoremstyle{remark}
\newtheorem*{remark}{Remark} % unnumbered
```

```
\begin{definition}
A graph is \emph{bipartite} if \dots
\end{definition}
\begin{lemma}\label{lem:odd}
$$$ is bipartite iff it has no odd cycle.
\end{lemma}
\begin{theorem}[Safe labeling]
Every tree admits a $k$-safe labeling
with span $n+k-1$.
\end{theorem}
\begin{proof}
A tree has no cycle, so by
Lemma~\ref{lem:odd} it is bipartite.
\end{proof}
```

Definition 4.1. A graph is *bipartite* if ...

Lemma 4.2. *G is bipartite iff it has no odd cycle.*

Theorem 4.3 (Safe labeling). *Every tree admits a k -safe labeling with span $n + k - 1$.*

Proof. A tree has no cycle, so by Lemma 4.2 it is bipartite. □

The `proof` environment ends with the `□` box automatically. The optional argument `[Safe labeling]` gives the theorem a name. (See chapter 18 for the meaning of Fact, Lemma, Theorem and Corollary.)

4.9 Useful maths extras

```
\[ \underbrace{1+\cdots+1}_{n \text{ times}}
\quad \overbrace{a+b}^{\text{sum}} \]
\[ x \xrightarrow{f} y \quad
A \stackrel{\text{def}}{=} B \]
\[ \boxed{E=mc^2} \]
```

$$\underbrace{1 + \cdots + 1}_{n \text{ times}} \quad \overbrace{a + b}^{\text{sum}}$$

$$x \xrightarrow{f} y \quad A \stackrel{\text{def}}{=} B$$

$$\boxed{E = mc^2}$$

Chapter 5

Tables

5.1 The tabular environment

```
\begin{tabular}{|l|c|r|}  
  \hline  
  Left & Centre & Right \\  
  \hline  
  a & b & c \\  
  longer text & x & 100 \\  
  \hline  
\end{tabular}
```

Left	Centre	Right
a	b	c
longer text	x	100

- The column specification in **braces** lists one letter per column: **l** left, **c** centre, **r** right, **p{3cm}** a paragraph column of fixed width (text wraps).
- **|** draws a vertical line; **||** a double line.
- **&** separates cells; **** ends a row.
- **\hline** is a full horizontal line; **\cline{2-3}** covers columns 2–3 only.
- ***{5}{c}** repeats: same as **ccccc**.
- **@{}** removes the padding at the edge; **@{--}** puts a dash between columns.

Quiz 1(h): five columns, first left, last right, others centred, only outer vertical lines

Answer: `{|lcccr|}` (B). Option A uses square brackets, which is the *optional* argument, not the column spec. Option D has no vertical lines at all. Option C uses `!`, which is not a column type.

Quiz 1(i): correct table syntax

`\begin{tabular}` is correct. `\start{table}` and `tables` do not exist. Note the difference: `tabular` builds the grid; `table` is the *floating* container that adds a caption and placement.

5.2 Professional tables with booktabs

Good tables have no vertical lines and only three horizontal rules.

```
\begin{tabular}{@{}lrr@{}}
\toprule
Algorithm & Time (ms) & Memory \\
\midrule
Dijkstra & 12.4 & 3.1 MB \\
Bellman--Ford & 48.9 & 2.8 MB \\
\bottomrule
\end{tabular}
```

Algorithm	Time (ms)	Memory
Dijkstra	12.4	3.1 MB
Bellman--Ford	48.9	2.8 MB

`\toprule`, `\midrule`, `\bottomrule` have correct thickness and spacing; `\cmidrule(1r){2-3}` is a partial rule.

5.3 Merging cells

```
\begin{tabular}{|c|c|c|}
\hline
\multicolumn{3}{|c|}{Header over 3} \\
\hline
\multirow{2}{*}{Tall} & a & b \\
\cline{2-3}
& c & d \\
\hline
\end{tabular}
```

Header over 3		
Tall	a	b
	c	d

- `\multicolumn{n}{spec}{text}`: span n columns with its own alignment and lines.
- `\multirow{n}{width}{text}` (package `multirow`): span n rows; `*` means natural width. Leave the cells below it empty.

5.4 Row height, column width and colour

```
\renewcommand{\arraystretch}{1.8}
\setlength{\tabcolsep}{12pt}
\begin{tabular}{|c|c|}
\hline 1 & 2 \\ \hline 3 & 4 \\ \hline
\end{tabular}
```

1	2
3	4

- `\arraystretch`: multiplies row height (default 1). Change it with `\renewcommand`, not `\setlength`.
- `\tabcolsep`: half the space between columns (a length, set with `\setlength`).
- Package `xcolor` with option `table`: `\rowcolor{gray!20}`, `\cellcolor{yellow}`, `\rowcolors{2}{gray!10}`

5.5 Tables that fit the width, and long tables

```
\begin{tabularx}{\linewidth}{|l|X|}
\hline
Term & Explanation that is long and wraps
automatically in the X column. \\
\hline
\end{tabularx}
```

Term	Explanation that is long and wraps automatically in the X column.
------	---

`tabularx` takes a total width, and `X` columns share the leftover space. `longtable` breaks across pages and repeats the header (`\endhead`). `\resizebox{\linewidth}{!}{...}` shrinks a wide table (use sparingly: it shrinks the font too).

5.6 The floating table environment

```
\begin{table}[htbp]
\centering
\caption{Running time of the algorithms.} % captions ABOVE tables
\label{tab:time}
\begin{tabular}{lr} ... \end{tabular}
\end{table}
As shown in Table~\ref{tab:time}, ...
```

Convention: table captions go *above* the table, figure captions *below* the figure.

5.7 Worked quiz answer: the blank beamer table

```
\renewcommand{\arraystretch}{2}
\begin{tabular}{|c|c|c|c|}
\hline
\phantom{000} & \phantom{000} & \phantom{000} & \phantom{000}\\ \hline
\phantom{000} & \phantom{000} & \phantom{000} & \phantom{000}\\ \hline
\phantom{000} & \phantom{000} & \phantom{000} & \phantom{000}\\ \hline
\end{tabular}\qqquad
\begin{tabular}{|c|c|c|c|}
\hline
1 & 2 & 3 & 4\\ \hline
11 & 22 & 33 & 44\\ \hline
111 & 222 & 333 & 444\\ \hline
\end{tabular}
```


1	2	3	4
11	22	33	44
111	222	333	444

Both tables are the same size: each cell reserves the width of “000” (the widest number), and `\arraystretch` doubles the row height. Alternatives: a fixed-width column `p{1cm}`, or `\makebox[1cm]{}` in each cell. In beamer you could also build one frame with `\only<2>{111}` or `\uncover<2>{111}` so the table appears blank on slide 1 and filled on slide 2 (`\uncover` reserves space; `\only` does not).

Chapter 6

Figures, Floats and Images

6.1 Including images

```
\usepackage{graphicx}
\graphicspath{{figures/}} % where to look
\includegraphics[width=0.6\linewidth]{graph.pdf}
\includegraphics[height=4cm]{photo.jpg}
\includegraphics[scale=0.5, angle=90]{plot.png}
\includegraphics[width=\linewidth, trim=1cm 0 1cm 0, clip]{shot.png}
```

Give only width *or* height to keep the aspect ratio. pdf_latex accepts PDF, PNG, JPG (and EPS converted automatically).

6.2 Vector vs raster graphics

	Vector (PDF, SVG, EPS, TikZ)	Raster (PNG, JPG)
Stored as	shapes, lines, text (mathematical)	a grid of pixels
Zooming	sharp at any size	blurs, shows pixels
Best for	diagrams, plots, charts, flowcharts, graphs	photos, screenshots, images with gradients
File size	small for diagrams	large at high resolution

Quiz 3(c)

Use **vector graphics** for anything made of lines, shapes and text (graph drawings, plots, architecture diagrams): they stay sharp when printed, zoomed or projected. Use **raster** for photographs, e.g. of an experimental setup, or screenshots.

Pie chart vs bar chart. A *pie chart* shows parts of a whole that add to 100%, with few categories: e.g. the share of CPU time spent in each phase of a program. A *bar chart* compares values across categories, or when values do not form a whole: e.g. the running time of five algorithms, or accuracy per dataset. Bar charts are also easier to read precisely because length is easier to judge than angle.

6.3 The figure environment and float placement

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.7\linewidth]{graph}
  \caption{A  $\$k\$$ -safe labeling with span 10.} % caption BELOW figures
  \label{fig:safe}
\end{figure}
```

A *float* is moved by LaTeX to a good position. The optional specifier lists allowed positions:

h	here, approximately where it appears in the source
t	top of a page
b	bottom of a page
p	a separate page with only floats
!	ignore some aesthetic restrictions (“try harder”)
H	exactly here, no floating (package float)

Quiz 1(f): are [htbp] and [pbth] identical?

Yes: the *order* of letters does not matter. LaTeX always tries the positions in the fixed order **h**, **t**, **b**, **p**, using only those you allowed. So [htbp], [pbth] and [thbp] all produce the same placement. Answer: (C) Both (A) and (B).

If figures pile up at the end, use `\clearpage` or `\FloatBarrier` (package `placeins`), give more options ([htbp]), or reduce figure size.

6.4 Sub-figures and side-by-side figures

```
\usepackage{subcaption}
\begin{figure}[htbp]
  \centering
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{a}
    \caption{Span 30.}\label{fig:a}
  \end{subfigure}\hfill
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{b}
    \caption{Span 10.}\label{fig:b}
  \end{subfigure}
  \caption{Two  $\$k\$$ -safe labelings for  $\$k=5\$$ .}\label{fig:both}
\end{figure}
Figure~\ref{fig:a} ... % prints 1a
```

Two independent figures side by side: use two `minipages` inside one `figure`, each with its own `\caption`.

6.5 Boxes and minipages

```
\begin{minipage}[t]{0.45\linewidth}
Left column text that wraps.
\end{minipage}\hfill
\begin{minipage}[t]{0.45\linewidth}
Right column text.
\end{minipage}

\fbbox{framed} \framebox[3cm][c]{fixed}
\makebox[3cm][r]{right in 3cm}
\parbox{3cm}{a paragraph box of 3cm}
\raisebox{1ex}{raised}
\rotatebox{30}{rotated}
\scalebox{1.5}{scaled}
```

Left column text that wraps.

Right column text.

framed

fixed

right in 3cm

a paragraph box raised

rotated

scaled

Chapter 7

Bibliography and Citations

7.1 Manual: thebibliography

As shown by Gallian~\cite{gallian} and in~\cite{west,gross} ...

```
\begin{thebibliography}{99}
\bibitem{gallian} J.~A. Gallian, ``A dynamic survey of graph labeling,''  

\emph{The Electronic Journal of Combinatorics}, vol.~16, pp.~1--308, 2013.
\bibitem{west} D.~B. West, \emph{Introduction to Graph Theory}, 2nd ed.  

Prentice Hall, 2001.
\end{thebibliography}
```

The argument {99} is the **widest label**: LaTeX uses it only to set the indentation of the list. It does not limit the number of items; {9} means “labels are one digit wide”, {99} “two digits”.

Quiz 1(j): maximum bibitems with \begin{thebibliography}{2}

The argument 2 is a *one-digit* widest label, so it is designed for up to **9** items (1–9) with correct alignment. The expected answer is (B) 9. (Technically more items still compile, but labels 10+ would overhang.)

7.2 BibTeX (the standard way)

Keep references in a .bib database file:

```
@article{gallian2013,
  author = {Gallian, Joseph A.},
  title = {A Dynamic Survey of Graph Labeling},
  journal = {The Electronic Journal of Combinatorics},
  volume = {16},
  pages = {1--308},
  year = {2013}
}
@inproceedings{rahaman2014,
  author = {Rahaman, M. S. and Eshan, T. A. and Rahman, M. S.},
  title = {Antibandwidth Problem for Itchy Caterpillars},
  booktitle = {Proc. of ICIEV},
```

```

    pages = {1--6},
    year = {2014}
}
@book{west2001,
  author = {West, Douglas B.},
  title = {Introduction to Graph Theory},
  edition = {2nd},
  publisher = {Prentice Hall},
  year = {2001}
}
@misc{overleaf, title={Overleaf}, howpublished={\url{https://overleaf.com}},
  note={Accessed 2026-09-27}}

```

Then in the document:

```

\cite{gallian2013} % [1]
\bibliographystyle{plain} % or IEEEtran, alpha, abbrv, unsrt, acm
\bibliography{refs} % refs.bib, without the extension

```

- Entry types: @article (journal), @inproceedings (conference), @book, @incollection (chapter), @phdthesis, @mastersthesis, @techreport, @misc.
- Only *cited* entries appear. \nocite{key} adds an uncited entry; \nocite{*} adds all.
- Protect capitals in titles with braces: {BUET}, {G}raph, or styles will lowercase them.
- Author names: Last, First and Last, First.

7.3 Styles and numbering schemes

Style	Label	Order
plain	[1]	alphabetical by author
unsrt	[1]	order of first citation (mention)
alpha	[Gal13]	alphabetical
abbrv	[1]	alphabetical, abbreviated names
IEEEtran	[1]	order of citation
plainnat (natbib)	(Gallian, 2013)	author–year

Quiz 1(d)

Reference items can be numbered in order of appearance in the list, in order of mention (unsrt), or alphabetically (plain, alpha). They can **not** be numbered randomly.

With natbib: \citet{key} gives “Gallian (2013)”, \citep{key} gives “(Gallian, 2013)”. biblatex with the biber backend is the modern alternative: \addbibresource{refs.bib} and \printbibliography.

7.4 Writing good references

From the flawed safelabeling.pdf in the quiz, references must be *consistent*: same author name format (J. A. Gallian vs Gallian J.), titles in quotes or italics consistently, complete page ranges (never “pp. ,”), venue, volume, year. BibTeX does this for you.

Chapter 8

Code and Algorithms

8.1 Source code with listings

```
\usepackage{listings}
\lstset{language=C, basicstyle=\ttfamily\small, numbers=left,
        keywordstyle=\color{blue}, commentstyle=\color{gray},
        frame=single, breaklines=true}

\begin{lstlisting}[language=Python, caption={Bubble sort}, label={lst:bs}]
def bubble(a):
    for i in range(len(a)):
        for j in range(len(a)-i-1):
            if a[j] > a[j+1]:
                a[j], a[j+1] = a[j+1], a[j]
\end{lstlisting}
\lstinputlisting[language=Java]{Main.java} % from a file
Inline: \lstinline|int x = 0;|
```

`minted` gives nicer highlighting (uses Python's `Pygments`) but needs `-shell-escape`.

8.2 Pseudocode with algorithm2e

```
\begin{algorithm}[H]
\caption{Two-colouring (bipartite test)}
\KwIn{A connected graph  $G=(V,E)$ }
\KwOut{A bipartition  $(P,Q)$ , or failure}
colour an arbitrary vertex  $s$  red\;
\ForEach{vertex  $u$  in BFS order from  $s$ }{
    \ForEach{neighbour  $v$  of  $u$ }{
        \uIf{ $v$  is uncoloured}{
            give  $v$  the opposite colour of  $u$ \;
        }
        \uElseIf{colour( $v$ ) = colour( $u$ )}{
            \Return failure (odd cycle)\;
        }
    }
}
\Return  $(P,Q)$ \;
```

```
\end{algorithm}
```

Algorithm 1: Two-colouring (bipartite test)

Input: A connected graph $G = (V, E)$

Output: A bipartition (P, Q) , or failure

```
1 colour an arbitrary vertex  $s$  red;
2 foreach vertex  $u$  in BFS order from  $s$  do
3   foreach neighbour  $v$  of  $u$  do
4     if  $v$  is uncoloured then
5       | give  $v$  the opposite colour of  $u$ ;
6     else if  $\text{colour}(v) = \text{colour}(u)$  then
7       | return failure (odd cycle);
8 return  $(P, Q)$ ;
```

Each statement ends with `\;`. Blocks use braces: `\If{cond}{...}`, `\eIf{c}{then}{else}`, `\While{c}{...}`, `\For{..}{..}`. The `algorithmicx/algpseudocode` package is the alternative style with `\State`, `\If{}` ... `\EndIf`.

```
\usepackage{algorithm, algpseudocode}
\begin{algorithm}
\caption{Euclid}
\begin{algorithmic}[1]
\Procedure{GCD}{$a$,  $b$ }
  \While{$b \neq 0$}
    \State  $(a, b) \gets (b, a \bmod b)$ 
  \EndWhile
  \State \Return  $a$ 
\EndProcedure
\end{algorithmic}
\end{algorithm}
```

Chapter 9

Custom Commands, Counters and Advanced LaTeX

9.1 Macros with `\newcommand`

```
\newcommand{\R}{\mathbb{R}}
\newcommand{\norm}[1]{\left\lVert #1 \right\rVert}
\newcommand{\pair}[2][x]{(#1, #2)}
$f:\R\to\R$, $\norm{v}$ and $\pair{y}$ and $\pair{a}{b}$
```

$f : \mathbb{R} \rightarrow \mathbb{R}, \|v\|$
 (x, y) and (a, b)

- Syntax: `\newcommand{\name}[number of args][default of first]{definition}`, with #1, #2... for arguments (up to 9).
- `\newcommand` errors if the name exists; `\renewcommand` redefines an existing command; `\providecommand` defines only if missing.
- Macros make notation consistent: change `\R` once, and every use changes.
- `\xspace` (package `xspace`) fixes the swallowed space after text macros.

9.2 New environments

```
\newenvironment{note}[1]
{\par\noindent\textbf{Note (#1):}\itshape}
{\par}
\begin{note}{exam}
Revise floats and bibliography.
\end{note}
```

Note (exam): *Revise floats and bibliography.*

`\newenvironment{name}[args]{begin code}{end code}`. Arguments can only be used in the *begin* code.

9.3 Counters

```
\newcounter{task}
\setcounter{task}{4}
\stepcounter{task}
Task \arabic{task}, \roman{task},
\Alph{task}. \\\
\addtocounter{task}{10}
Now \thetask.
```

Task 5, v, E.
Now 15.

Built-in counters: page, section, equation, figure, table, footnote, enumi. `\refstepcounter` increments and makes the value referable by `\label`. `\renewcommand{\thefigure}{\arabic{section}.\arabic{figure}}` changes how a counter prints.

9.4 Lengths

```
\newlength{\mylen}
\setlength{\mylen}{2cm}
\addtolength{\mylen}{5mm}
\settowidth{\mylen}{Longest label} % measure text
\setlength{\parindent}{0pt}
\setlength{\parskip}{1ex plus 0.5ex minus 0.2ex} % "rubber" length
```

A *rubber* (glue) length can stretch and shrink, which lets LaTeX justify lines and pages. `\stretch{1}` and `\fill` are infinitely stretchable.

9.5 Conditionals and loops

```
\usepackage{ifthen}
\ifthenelse{\equal{\mode}{draft}}{Draft}{Final}

\newif\ifanswers \answerstrue % toggle solutions on/off
\ifanswers Solution: ... \fi

\usepackage{pgffor}
\foreach \i in {1,...,5}{Item \i. } % also works outside TikZ
```

The `\newif` trick is very useful for making a question paper and its answer key from one source.

9.6 Useful tools and packages

- `todonotes`: `\todo{fix this}` in the margin.
- `siunitx`: `\num{12345.678}`, `\SI{9.81}{\metre\per\second\squared}`, and column type `S` aligns decimals in tables.

- `tcolorbox`: coloured boxes (used in this guide).
- `hyperref` options: `\hypersetup{colorlinks=true, linkcolor=blue}, \autoref`.
- `babel/polyglossia`: language hyphenation; Bangla with XeLaTeX + `fontspec` and a Bangla font such as Kalpurush.
- `makeidx`: `\index{graph}` and `\printindex` build a book index.
- `glossaries`: acronyms (`\gls{cpu}` expands the first time).

Book elements from the syllabus

Preparing theses and books: **abstract** (`abstract`), **preface** (`\chapter*{Preface}` in `\frontmatter`), **contents** (`\tableofcontents`), **bibliography** (BibTeX) and **index** (`makeidx`).

Part II

TikZ: Drawing in LaTeX

Chapter 10

TikZ Fundamentals

10.1 Setting up

TikZ (“TikZ ist kein Zeichenprogramm”) draws vector graphics from code, so diagrams share the document’s fonts and stay sharp.

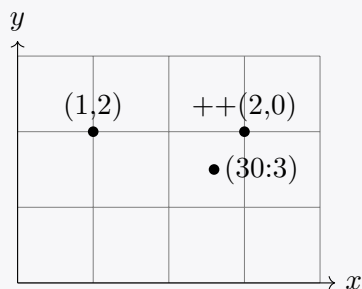
```
\usepackage{tikz}
\usetikzlibrary{arrows.meta, positioning, shapes.geometric, calc, decorations.
    pathreplacing}
...
\begin{tikzpicture}[scale=1, options]
    ... drawing commands, each ending with ;
\end{tikzpicture}
```

Common mistake

Every TikZ command ends with a **semicolon**. Forgetting it is the most common TikZ error.

10.2 Coordinates

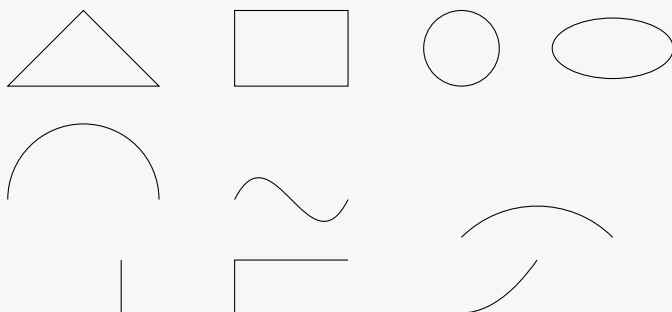
```
\begin{tikzpicture}
\draw[help lines] (0,0) grid (4,3);
\draw[->] (0,0) -- (4.2,0) node[right]{$x$};
\draw[->] (0,0) -- (0,3.2) node[above]{$y$};
\fill (1,2) circle (2pt) node[above]{(1,2)};
\fill (30:3) circle (2pt) node[right]{(30:3)};
\fill (1,2) ++(2,0) circle (2pt)
    node[above]{++(2,0)};
\end{tikzpicture}
```



- **Cartesian** (x,y): default unit is **1 cm**. Units can be given: (2pt,1mm).
- **Polar** (angle:radius): (30:3) is 3 cm at 30°.
- **Relative** ++(dx,dy): move from the last point *and* update it. +(dx,dy) moves without updating.
- **Named** (A): a node or coordinate defined by name.

10.3 Paths: lines, shapes, curves

```
\begin{tikzpicture}
\draw (0,0) -- (1,1) -- (2,0) -- cycle;
\draw (3,0) rectangle (4.5,1);
\draw (6,0.5) circle (0.5);
\draw (8,0.5) ellipse (0.8 and 0.4);
\draw (0,-1.5) arc (180:0:1);
\draw (3,-1.5) .. controls (3.5,-0.5)
and (4,-2.5) .. (4.5,-1.5);
\draw (6,-2) to[bend left=45] (8,-2);
\draw (0,-3) -| (1.5,-2.3);
\draw (3,-3) |- (4.5,-2.3);
\draw (6,-3) parabola (7,-2.3);
\end{tikzpicture}
```



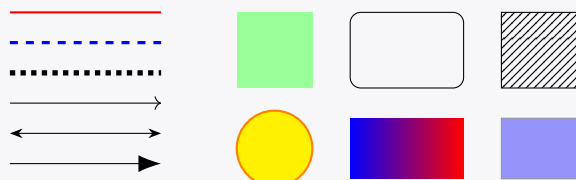
Operation	Meaning
--	straight line
-- cycle	close the path back to the start (clean corner)
- , -	horizontal then vertical, or vertical then horizontal
rectangle (x,y)	rectangle from the current point to the opposite corner
circle (r), ellipse (a and b)	circle, ellipse at the current point
arc (start:end:radius)	arc starting at the current point
.. controls (a) and (b) ..	Bézier curve
to[bend left=30], to[out=90,in=180]	curved connections
grid (x,y)	grid lines, <code>step=0.5</code> changes spacing

10.4 Drawing commands and options

```

\begin{tikzpicture}
\draw[red, thick] (0,0) -- (2,0);
\draw[blue, dashed, very thick] (0,-0.4) -- (2,-0.4);
\draw[dotted, line width=2pt] (0,-0.8) -- (2,-0.8);
\draw[->] (0,-1.2) -- (2,-1.2);
\draw[<->, >=Stealth] (0,-1.6) -- (2,-1.6);
\draw[-{Latex[length=3mm]}] (0,-2) -- (2,-2);
\fill[green!40] (3,0) rectangle (4,-1);
\filldraw[fill=yellow, draw=orange, thick]
  (3.5,-1.8) circle (0.5);
\draw[rounded corners] (4.5,0) rectangle (6,-1);
\shade[left color=blue, right color=red]
  (4.5,-1.4) rectangle (6,-2.2);
\draw[pattern=north east lines] (6.5,0) rectangle (7.5,-1);
\draw[opacity=0.4, fill=blue] (6.5,-1.4) rectangle (7.5,-2.2);
\end{tikzpicture}

```



- `\draw` strokes, `\fill` fills, `\filldraw` does both, `\shade` fills with a gradient, `\path` does nothing visible (used for coordinates and nodes), `\clip` restricts later drawing.
- Thickness: `ultra thin`, `very thin`, `thin`, `semithick`, `thick`, `very thick`, `ultra thick` or `line width=2pt`.
- Dashes: `dashed`, `densely dashed`, `loosely dashed`, `dotted`, `dash dot`.
- Arrows: `->`, `<-`, `<->`, `|-|`; tips from `arrows.meta`: `Stealth`, `Latex`, `Triangle`. Set globally with `>=Stealth`.
- Colours: any `xcolor` colour: `red!50!blue`; `opacity`, `fill opacity`.

Practice question: two arrowheads

`[<->]` puts arrowheads at both ends, `[->]` at the end, `[<-]` at the start.

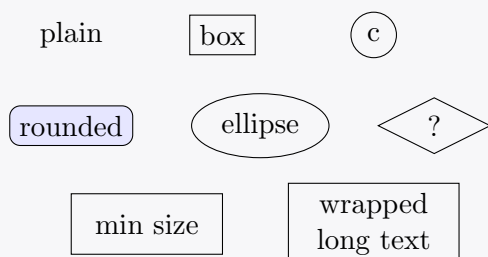
Chapter 11

Nodes, Positioning and Styles

11.1 Nodes

A node is a piece of text with an optional shape, placed on a path.

```
\begin{tikzpicture}
\node at (0,0) {plain};
\node[draw] at (2,0) {box};
\node[draw, circle] at (4,0) {c};
\node[draw, rounded corners, fill=blue!10]
  at (0,-1.2) {rounded};
\node[draw, ellipse] at (2.5,-1.2) {ellipse};
\node[draw, diamond, aspect=2] at (4.8,-1.2) {?};
\node[draw, minimum width=2cm,
  minimum height=8mm] at (1,-2.5) {min size};
\node[draw, text width=2cm, align=center]
  at (4,-2.5) {wrapped long text};
\end{tikzpicture}
```



Node options: `draw`, `fill`, `shape` (rectangle default, circle, ellipse, diamond, trapezium, regular polygon; the last few from `shapes.geometric`), `minimum size`, `inner sep` (padding), `outer sep`, `text width`, `align`, `font=\small`, `text=red`.

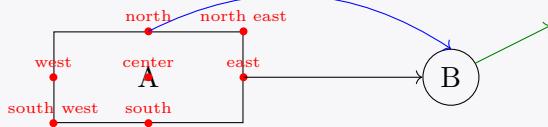
11.2 Named nodes and anchors

```
\begin{tikzpicture}
\node[draw, minimum width=2.5cm,
  minimum height=1.2cm] (A) {A};
\foreach \a in {north, south, east, west,
```

```

north east, south west, center}
\fill[red] (A.\a) circle (1.5pt)
  node[font=\tiny, above] {\a};
\node[draw, circle] (B) at (4,0) {B};
\draw[->] (A) -- (B);
\draw[->, blue] (A.north) to[bend left] (B.north);
\draw[->, green!50!black] (B.30) -- ++(1,0.5);
\end{tikzpicture}

```



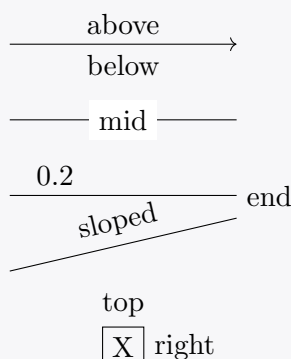
`\node (name) at (x,y) {text};`. Lines to a node stop at its border automatically. Anchors: north, south, east, west, north east, ..., center, and angles like (B.30). `\coordinate(P) at (1,2);` names a point without drawing anything.

11.3 Labels on paths

```

\begin{tikzpicture}
\draw[->] (0,0) -- node[above]{above}
  node[below]{below} (3,0);
\draw (0,-1) -- node[midway, fill=white]{mid}
  (3,-1);
\draw (0,-2) -- (3,-2)
  node[pos=0.2, above]{0.2}
  node[at end, right]{end};
\draw (0,-3) -- node[sloped, above]{sloped}
  (3,-2.3);
\node[draw, label=above:top,
  label=right:{right}] at (1.5,-4) {X};
\end{tikzpicture}

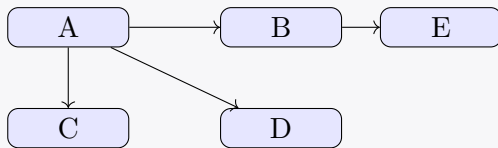
```



A node placed *inside* a path (after `--`) sits at its midpoint; `pos=0..1` chooses a point along it; `sloped` rotates with the line; `auto` puts labels on the outside of curves.

11.4 Relative positioning

```
\begin{tikzpicture}[node distance=8mm and 12mm,
  box/.style={draw, rounded corners,
    fill=blue!10, minimum width=16mm}]
\node[box] (a) {A};
\node[box, right=of a] (b) {B};
\node[box, below=of a] (c) {C};
\node[box, below right=of a] (d) {D};
\node[box, right=5mm of b] (e) {E};
\draw[->] (a) -- (b);
\draw[->] (a) -- (c);
\draw[->] (a) -- (d);
\draw[->] (b) -- (e);
\end{tikzpicture}
```



Needs `\usetikzlibrary{positioning}`. `right=of a` measures node distance between the *borders*; `right=5mm of b` sets a specific gap; `node distance=V` and `H` sets vertical and horizontal gaps.

11.5 Styles

```
% Local: in the tikzpicture options
\begin{tikzpicture}[vertex/.style={circle, draw, fill=white, inner sep=2pt},
  edge/.style={thick, ->}]
% Global: in the preamble, for all pictures
\tikzset{every node/.style={font=\small},
  process/.style={rectangle, draw, rounded corners, fill=orange!20}}
% Style with an argument
\tikzset{col/.style={fill=#1!20, draw=#1}} % use as [col=red]
```

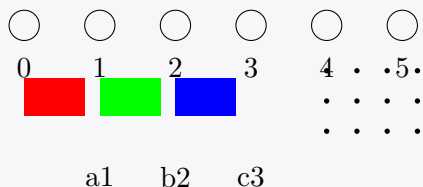
Styles are to TikZ what `\newcommand` is to LaTeX: define once, keep every node consistent. `every node/.style`, `every path/.style` apply automatically.

Chapter 12

TikZ Techniques

12.1 Loops with `\foreach`

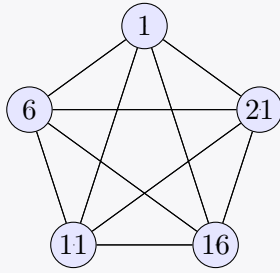
```
\begin{tikzpicture}
\foreach \x in {0,1,...,5}
  \draw (\x,0) circle (0.2) node[below=3mm]{\x};
\foreach \x/\c in {0/red, 1/green, 2/blue}
  \fill[\c] (\x,-1.2) rectangle ++(0.8,0.5);
\foreach \x in {0,...,3}
  \foreach \y in {0,...,2}
    \fill (4+\x*0.4, -1.4+\y*0.4) circle (1pt);
\foreach \i [count=\n] in {a,b,c}
  \node at (\n,-2) {\i\n};
\end{tikzpicture}
```



`{1,...,5}` fills in the range; `{0,0.5,...,2}` infers the step. `\x/\y in {a/b, c/d}` iterates over pairs. `[count=\n]` numbers items. Arithmetic works inside coordinates: `(\x*0.4, 2)`.

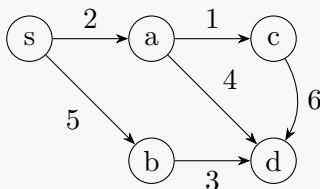
12.2 Drawing graphs (vertices and edges)

```
\begin{tikzpicture}[v/.style={circle, draw,
  fill=blue!10, minimum size=6mm, inner sep=0}]
\foreach \i/\lab in {0/1,1/6,2/11,3/16,4/21}
  \node[v] (n\i) at (90+72*\i:1.6) {\lab};
\foreach \i in {0,...,4}
  \foreach \j in {0,...,4}
    \draw (n\i) -- (n\j);
\end{tikzpicture}
```



This is K_5 with a 5-safe labeling of span 21 (from the safelabeling paper, Fig. 2a). Vertices are placed in polar coordinates on a circle; the double loop joins every pair of vertices (each edge is drawn twice, which looks identical).

```
\begin{tikzpicture}[v/.style={circle,draw,
  minimum size=6mm,inner sep=0},>=Stealth]
\node[v] (s) {s};
\node[v, right=of s] (a) {a};
\node[v, below=of a] (b) {b};
\node[v, right=of a] (c) {c};
\node[v, right=of b] (d) {d};
\draw[->] (s) -- node[above]{2} (a);
\draw[->] (s) -- node[below left]{5} (b);
\draw[->] (a) -- node[above]{1} (c);
\draw[->] (b) -- node[below]{3} (d);
\draw[->] (a) -- node[above right]{4} (d);
\draw[->] (c) to[bend left] node[right]{6} (d);
\end{tikzpicture}
```



A weighted directed graph: edge labels are nodes placed on the edge.

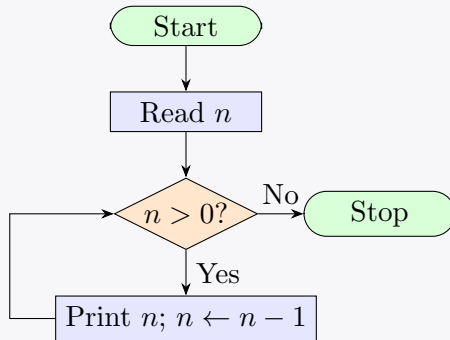
12.3 Flowcharts

```
\begin{tikzpicture}[node distance=6mm,>=Stealth,
  term/.style={draw, rounded corners=3mm,
    fill=green!15, minimum width=2cm},
  proc/.style={draw, fill=blue!10,
    minimum width=2cm},
  dec/.style={draw, diamond, aspect=2,
    fill=orange!20, inner sep=1pt}]
\node[term] (start) {Start};
\node[proc, below=of start] (read) {Read $n$};
\node[dec, below=of read] (chk) {$n>0$?};
\node[proc, below=of chk] (out)
  {Print $n$; $\textcolor{blue}{n}\text{gets } n-1$};
```

```

\node[term, right=of chk] (stop) {Stop};
\draw[->] (start) -- (read);
\draw[->] (read) -- (chk);
\draw[->] (chk) -- node[right]{Yes} (out);
\draw[->] (chk) -- node[above]{No} (stop);
\draw[->] (out.west) -- ++(-0.6,0) |- (chk.west);
\end{tikzpicture}

```



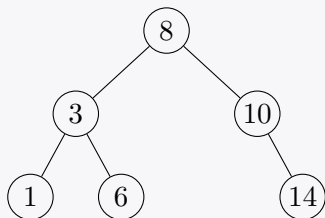
The loop-back arrow uses `++(-0.6,0) |- (chk.west)`: go left, then vertical, then horizontal into the diamond.

12.4 Trees

```

\begin{tikzpicture}[level distance=11mm,
  level 1/.style={sibling distance=24mm},
  level 2/.style={sibling distance=12mm},
  every node/.style={circle, draw,
    minimum size=6mm, inner sep=0}]
\node {8}
  child { node {3}
    child { node {1} }
    child { node {6} } }
  child { node {10}
    child[missing]
    child { node {14} } };
\end{tikzpicture}

```



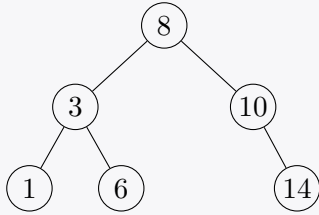
`child{...}` builds a tree; `child[missing]` keeps an empty slot so 14 is a right child. For bigger trees the `forest` package is simpler:

```

\begin{forest}
for tree={circle, draw, minimum size=6mm,
  inner sep=0, s sep=6mm}

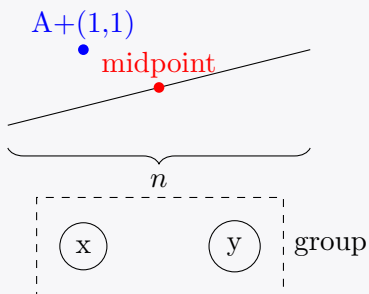
```

```
[8 [3 [1] [6]] [10 [,phantom] [14]]]
\end{forest}
```



12.5 The calc library, braces and fitting

```
\begin{tikzpicture}
\coordinate (A) at (0,0);
\coordinate (B) at (4,1);
\draw (A) -- (B);
\fill[red] ($(A)!0.5!(B)$) circle (2pt)
  node[above]{midpoint};
\fill[blue] ($(A)+(1,1)$) circle (2pt)
  node[above]{A+(1,1)};
\draw[decorate, decoration={brace,
  amplitude=6pt, mirror}]
  (0,-0.3) -- (4,-0.3) node[midway,below=6pt]{$n$};
\node[draw, circle] (x) at (1,-1.6) {x};
\node[draw, circle] (y) at (3,-1.6) {y};
\node[draw, dashed, fit=(x) (y), inner sep=3mm,
  label=right:group] {};
\end{tikzpicture}
```



- $\$(A)+(1,1)\$$: coordinate arithmetic (the $\$$ here is TikZ syntax, not maths).
- $\$(A)!0.5!(B)\$$: the point halfway from A to B; $!0.25!$ is a quarter of the way.
- `decorations.pathreplacing`: braces; `mirror` flips the brace.
- `fit`: a node that surrounds other nodes.

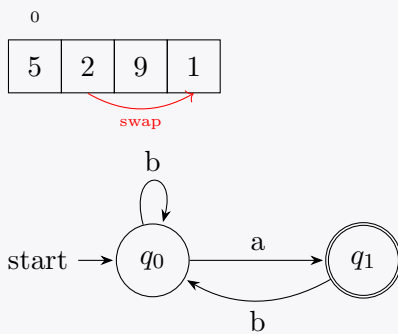
12.6 Matrices of nodes, layers and automata

```

\begin{tikzpicture}
\matrix (m) [matrix of nodes, nodes={draw,
  minimum size=7mm}, column sep=-\pgflinewidth,
  row sep=-\pgflinewidth]
{ 5 & 2 & 9 & 1 \\ };
\node[above=1mm of m-1-1]{\tiny 0};
\draw[->, red] (m-1-2.south) to[bend right]
  node[below]{\tiny swap} (m-1-4.south);
\end{tikzpicture}

\begin{tikzpicture}[shorten >=1pt,
  node distance=18mm, auto, >=Stealth]
\node[state, initial] (q0) {$q_0$};
\node[state, accepting, right=of q0] (q1) {$q_1$};
\path[->] (q0) edge node{a} (q1)
  (q0) edge[loop above] node{b} ()
  (q1) edge[bend left] node{b} (q0);
\end{tikzpicture}

```



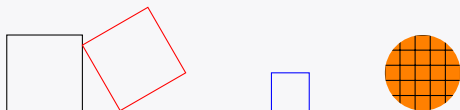
The `matrix` library draws arrays (sorting, memory). Cells are named `m-row-col`. The `automata` library gives `state`, `initial`, `accepting` for finite automata. `\path ... edge` draws edges between existing nodes; `loop above` draws self-loops.

12.7 Transformations, clipping and backgrounds

```

\begin{tikzpicture}
\draw (0,0) rectangle (1,1);
\draw[red, shift={(1.5,0)}, rotate=30]
  (0,0) rectangle (1,1);
\draw[blue, xshift=3.5cm, scale=0.5]
  (0,0) rectangle (1,1);
\begin{scope}[xshift=5cm]
\clip (0.5,0.5) circle (0.5);
\fill[orange] (0,0) rectangle (1,1);
\draw[step=0.2] (0,0) grid (1,1);
\end{scope}
\end{tikzpicture}

```



`scope` applies options to a group of commands. Options: `shift`, `xshift`, `yshift`, `rotate`, `scale`, `xscale`, `yscale`. `backgrounds` library: `\begin{scope}[on background layer]` draws behind nodes.

12.8 TikZ inside figures and scaling

```
\begin{figure}[htbp]
  \centering
  \begin{tikzpicture}[scale=0.8, every node/.style={transform shape}]
    ...
  \end{tikzpicture}
  \caption{A flowchart drawn with TikZ.}\label{fig:flow}
\end{figure}
```

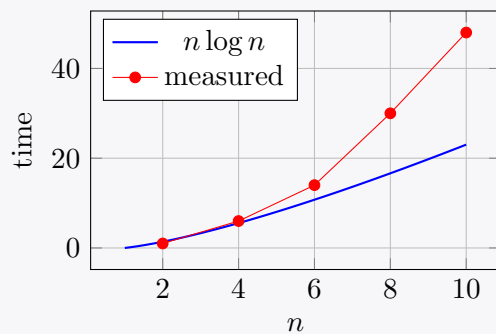
`scale` scales coordinates but *not* text or line widths; add `transform shape` to scale nodes too, or wrap the picture in `\resizebox{\linewidth}{!}{...}`. Draw complex figures in a separate file and `\input` them.

Chapter 13

Plots with PGFPlots

13.1 Function and data plots

```
\begin{tikzpicture}
\begin{axis}[width=7cm, height=5cm,
  xlabel={ $n$ }, ylabel={time},
  legend pos=north west, grid=major]
\addplot[blue, thick, domain=1:10,
  samples=50] {x*ln(x)};
\addlegendentry{{ $n \log n$ }}
\addplot[red, mark=*, coordinates
  {(2,1) (4,6) (6,14) (8,30) (10,48)}];
\addlegendentry{measured}
\end{axis}
\end{tikzpicture}
```



`\addplot {expr}` plots a function over domain; `coordinates {...}` plots data; `table{file.csv}` reads a file. Axis options: `xmin`, `xmax`, `ymin`, `ymax`, `xmode=log`, `title`, `legend pos`, `grid`.

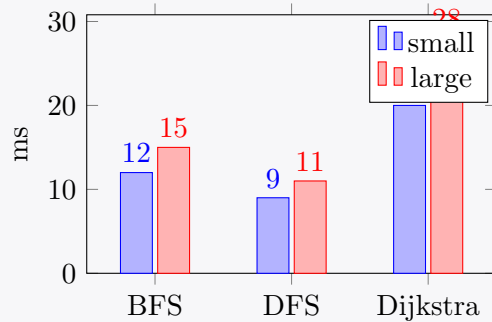
13.2 Bar charts

```
\begin{tikzpicture}
\begin{axis}[ybar, width=7cm, height=5cm,
  symbolic x coords={BFS,DFS,Dijkstra},
  xtick=data, ylabel={ms},
  nodes near coords, ymin=0,
```

```

    enlarge x limits=0.25, bar width=12pt]
\addplot coordinates
  {(BFS,12) (DFS,9) (Dijkstra,20)};
\addplot coordinates
  {(BFS,15) (DFS,11) (Dijkstra,28)};
\legend{small, large}
\end{axis}
\end{tikzpicture}

```

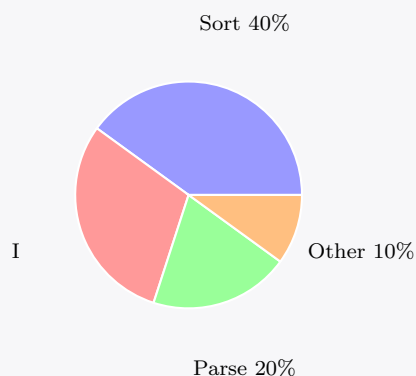


13.3 Pie chart (plain TikZ)

```

\begin{tikzpicture}
\def\r{1.5}
\foreach \s/\e/\c/\l in {0/144/blue!40/Sort 40\%,
  144/252/red!40/I 30\%,
  252/324/green!40/Parse 20\%,
  324/360/orange!50/Other 10\%}{
  \fill[\c, draw=white, thick] (0,0) --
    (\s:\r) arc (\s:\e:\r) -- cycle;
  \node at ({(\s+\e)/2}:\r+0.9)
    {\scriptsize \l};
}
\end{tikzpicture}

```



Each slice is a triangle-and-arc path from the centre; angles are percentage $\times 3.6$. The `pgf-pie` package automates this. Remember the rule from the quiz: pie for parts of a whole, bar for comparing quantities.

Part III

Beamer: Presentations in LaTeX

Chapter 14

Beamer Fundamentals

14.1 Frames and slides

Beamer is a document class for presentations. Its key idea is the difference between a **frame** and a **slide**.

- A **frame** is what you write in the source: one `\begin{frame} ... \end{frame}` block, one logical page of content.
- A **slide** is one page of the output PDF. One frame can produce *several* slides when you use overlays (`\pause`, `\only`, `\uncover`, `<2->`), so content appears step by step.

Quiz 3(d): how do slides and frames differ? Give an example.

A frame is the source unit; a slide is a PDF page. This one frame produces **three** slides:

```
\begin{frame}{Example}
  \begin{itemize}
    \item First point % slide 1
    \pause
    \item Second point % slide 2
    \pause
    \item Third point % slide 3
  \end{itemize}
\end{frame}
```

Visible from a distance: use large fonts (at least 20–24 pt on screen, few words per slide); keep high contrast between text and background (dark on light or light on dark); use simple sans-serif fonts; thick lines and big labels in figures; no more than about 5–6 bullet points per slide.

14.2 A complete beamer document

```
\documentclass[aspectratio=169]{beamer} % 16:9; default is 4:3
\usetheme{Madrid} % overall layout
\usecolortheme{beaver} % colours
\usefonttheme{professionalfonts} % fonts
\setbeamertemplate{navigation symbols}{} % hide the little icons
```

```

\title[Short title]{Safe Labeling of Graphs} % [short] goes in footline
\subtitle{A Beamer Demo}
\author[Student]{Your Name}
\institute[BUET]{Department of CSE, BUET}
\date{\today}

\begin{document}
\begin{frame}
  \titlepage % title slide
\end{frame}

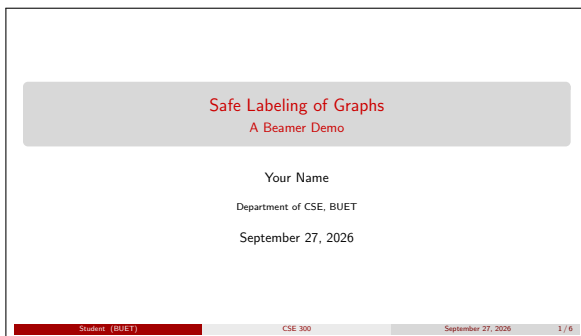
\begin{frame}{Outline}
  \tableofcontents % built from \section commands
\end{frame}

\section{Introduction} % sections go OUTSIDE frames
\begin{frame}{Frame title}{Optional frame subtitle}
  Content here.
\end{frame}
\end{document}

```

Frame title: either `\begin{frame}{Title}` or `\frametitle{Title}` inside the frame. `\section` and `\subsection` go *between* frames and build the outline and navigation bar.

Below are pages of this exact demo compiled (each box is one *slide*):



14.3 Frame options

```

\begin{frame}[fragile]{Code} % needed for verbatim / lstlisting inside
\begin{frame}[plain] % no header/footer (for big images)
\begin{frame}[allowframebreaks]{References} % long content over several slides
\begin{frame}[t] % align content to top (c = centre default, b = bottom)
\begin{frame}[shrink=10] % shrink content (last resort)
\begin{frame}[label=main] % name it; \againframe{main} repeats it

```

Common mistake

`[fragile]` is required whenever the frame contains `verbatim`, `\verb` or `lstlisting`; otherwise you get strange errors.

Chapter 15

Overlays: Building Slides Step by Step

15.1 `\pause`

`\pause` shows everything up to it on one slide, and the rest on later slides.



15.2 Overlay specifications `<...>`

Many commands accept an overlay specification that says on which slides they act:

Spec	Meaning
<code><2></code>	only slide 2
<code><2-></code>	slide 2 onwards
<code><-3></code>	slides 1 to 3
<code><2-4></code>	slides 2, 3, 4
<code><1,3,5></code>	slides 1, 3 and 5
<code><+-></code>	“next slide onwards” (auto-incrementing)

```
\begin{itemize}
  \item<1-> Always visible
  \item<2-> Appears on slide 2
  \item<3-> \alert<3>{Highlighted on slide 3}
\end{itemize}
\only<2>{Text only on slide 2 (no space kept).}
\uncover<3>{Text uncovered on slide 3 (space reserved).}

\begin{itemize}[<+->] % every item appears one by one
  \item A \item B \item C
\end{itemize}
```

Overlays: \only, \uncover, \alert

- Always visible

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Overlays: \only, \uncover, \alert

- Always visible
- Appears on slide 2

Text only on slide 2 (no space kept).

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Overlays: \only, \uncover, \alert

- Always visible
- Appears on slide 2
- Highlighted on slide 3

Text uncovered on slide 3 (space reserved).

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15.3 \only vs \uncover vs \visible vs \onslide

Command	Behaviour on slides where it is hidden
<code>\only<2>{..}</code>	content is removed ; no space reserved, so the layout can jump
<code>\uncover<2>{..}</code>	content is invisible (or transparent) but space is reserved
<code>\visible<2>{..}</code>	like <code>\uncover</code> , but always fully invisible
<code>\invisible<2>{..}</code>	opposite of <code>\visible</code>
<code>\onslide<2>-></code>	declaration: applies to everything after it
<code>\alert<2>{..}</code>	highlights (red by default) on those slides
<code>\temporal<2>{before}{during}{after}</code>	three versions for before/on/after
<code>overprint, onlyenv</code>	replace content in the same area

`\setbeamercovered{transparent}` shows hidden `\uncover`/`\pause` content faintly instead of hiding it.

15.4 Overlays in TikZ

With beamer, TikZ commands accept overlays directly:

```

\begin{tikzpicture}
  \node[v] (a) {a}; \node[v, right=of a] (b) {b};
  \node<2->[v, below=of a] (c) {c}; % appears on slide 2
  \draw (a) -- (b);
  \draw<2-> (a) -- (c) (b) -- (c); % edges appear on slide 2
  \node<3>[red] at (1,1) {done}; % only slide 3
\end{tikzpicture}

```

Columns and TikZ step by step

- Left column
- Text here



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Columns and TikZ step by step

- Left column
- Text here



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This is ideal for explaining algorithms: show the graph, then highlight each step.

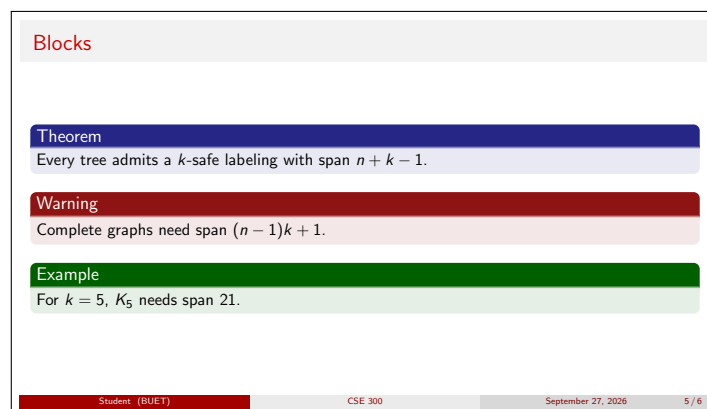
58

Chapter 16

Beamer Layout, Themes and Content

16.1 Blocks

```
\begin{block}{Theorem}
  Every tree admits a  $k$ -safe labeling with span  $n+k-1$ .
\end{block}
\begin{alertblock}{Warning} ... \end{alertblock} % red
\begin{exampleblock}{Example} ... \end{exampleblock} % green
% Also: theorem, lemma, definition, proof, example environments are predefined
```



16.2 Columns

```
\begin{columns}[T] % T = align tops
  \begin{column}{0.5\textwidth}
    Text or a list
  \end{column}
  \begin{column}{0.5\textwidth}
    \includegraphics[width=\linewidth]{fig}
  \end{column}
\end{columns}
% short form:
\begin{columns} \column{0.4\textwidth} ... \column{0.6\textwidth} ... \end{columns}
```

16.3 Themes

- **Presentation themes** (`\usetheme`): default, Madrid, Berlin, Warsaw, CambridgeUS, Copenhagen, AnnArbor, Boadilla, Frankfurt, Singapore, metropolis (modern, clean).
- **Colour themes** (`\usecolortheme`): beaver, crane, dolphin, whale, seahorse, orchid, rose.
- **Font themes** (`\usefonttheme`): serif, structurebold, professionalfonts.
- **Inner/outer themes**: `\useinnertheme{circles}`, `\useoutertheme{infolines}`.

Fine control:

```
\setbeamercolor{frametitle}{fg=white, bg=blue!70!black}
\setbeamerfont{frametitle}{size=\Large, series=\bfseries}
\setbeamertemplate{footline}[frame number]
\setbeamertemplate{itemize items}[circle]
\definecolor{buetblue}{RGB}{0,82,147}
\usecolortheme[named=buetblue]{structure}
\logo{\includegraphics[height=8mm]{buet-logo}}
```

16.4 Other useful beamer features

- **Section outline slides**: `\AtBeginSection[]{\begin{frame}{Outline}\tableofcontents[currentsection]}` shows the outline with the current section highlighted.
- **Appendix / backup slides**: `\appendix` after the last main frame; backup slides do not count in the total.
- **Handouts**: `\documentclass[handout]{beamer}` collapses all overlays into one slide per frame (for printing).
- **Speaker notes**: `\note{...}` with `\setbeameroption{show notes on second screen}`.
- **Hyperlinks**: `\hyperlink{main}{\beamergotobutton{Go}}` to jump to a labelled frame.
- **Bibliography in beamer**: use a frame with `[allowframebreaks]` and `\begin{thebibliography}{9}`.
- **Transitions**: `\transfade`, `\transdissolve` (use rarely).

16.5 Beamer tables and figures

`figure` and `table` do not float in beamer (there is nowhere to float to) and are not numbered by default; `\caption` prints “Figure:” / “Table:”. That is why the quiz’s figure shows “Table: Table for Quiz Question 3(a)”. Add `\setbeamertemplate{caption}[numbered]` to number them.

Chapter 17

Presentation Skills

17.1 Designing good slides

- **One idea per slide.** The title should state the point, not just the topic (“Trees need span $n + k - 1$ ” rather than “Results”).
- **Few words.** Slides support your talk; they are not the script. Aim for keywords, not full sentences, and at most 5–6 bullets.
- **Visible from the back of the room.** Large font, high contrast, sans-serif, thick lines. Never paste a dense table: show the 2–3 numbers that matter.
- **Figures over text.** A diagram or plot explains faster; use vector graphics so they stay sharp on the projector.
- **Consistent design.** One theme, one colour scheme, consistent fonts and positions.
- **Build complex ideas step by step** with overlays instead of showing everything at once.
- **Number your slides** so the audience can ask “go back to slide 7”.

17.2 Timing

Quiz 1(c): good presentation practice

Keep a small number of slides but assign enough time to each slide (answer A). A common rule of thumb is about 1–2 minutes per slide: a 15-minute talk needs roughly 10–12 slides. Rushing through many slides loses the audience.

17.3 Structure of a research talk

1. Title slide (title, authors, affiliation, venue).
2. Motivation: why should the audience care? Start with an example.
3. Problem definition, with a small example figure.
4. Previous work (brief) and the gap.
5. Your contributions (bullet list).

6. Main idea and intuition; only then key details.
7. Results: one or two plots, clearly labelled.
8. Conclusion and future work; the one thing to remember.
9. “Thank you / Questions?” slide; keep backup slides for expected questions.

17.4 Delivery

Face the audience, not the screen. Speak slowly and clearly; rehearse aloud with a timer at least three times. Point to what you are talking about. Anticipate questions and prepare backup slides. Repeat each question before answering it. Finish on time; running over is disrespectful to the next speaker.

Part IV

Technical Writing (CSE 300 Slides)

Chapter 18

Issues of Technical Writing

18.1 The course syllabus

CSE 300 (Md. Saidur Rahman, BUET) covers:

- Issues of technical writing and oral presentation in computer science and engineering.
- Writing styles of definitions, propositions, theorems and proofs.
- Preparation of reports, research papers, theses and books: abstract, preface, content, bibliography and index.
- Writing book reviews and referee reports.
- Writing tools: LaTeX, diagram drawing software, presentation tools.

Parts I–III of this guide cover the tools; this part covers the writing itself.

18.2 Why do we need writing?

- **To express our ideas.** An idea that is not written down cannot be checked, used or credited.
- **Writing improves understanding.** Writing a good mathematical explanation forces you to understand every step; gaps you did not notice in your head become obvious on paper.
- **Careful writing is more likely to be correct.** Putting an idea on paper requires careful thought and attention, so mathematics that is written clearly and carefully is more likely to be right.

Specifically, as a computer scientist you will write:

- research papers to publish your results;
- theses or dissertations to fulfil degree requirements;
- technical textbooks for students;
- review reports (referee reports on others' papers, book reviews);
- technical study reports, project reports, documentation, proposals.

18.3 Who is your audience?

Your peers: people with a similar background, who are intelligent but do not know your specific work. Imagine someone you are explaining your ideas to, and consider:

- what questions they might ask;
- what confusions might arise;
- which details you need to bring out and explain.

Writing for yourself leads to skipped steps; writing for a complete beginner leads to tedium. Aim at a classmate who missed your project meetings.

Chapter 19

Aspects of Technical Writing: Worked Examples

The slides teach four principles through bad examples. Understand *why* each is bad.

19.1 Say what you mean; mean what you say

Bad

As a valued customer of XYZ company, your call is very important to us.

What is wrong? This is a *dangling modifier*. The opening phrase “As a valued customer” attaches grammatically to the subject of the main clause, which is “your call”. So the sentence literally says that *your call* is a valued customer.

Better

You are a valued customer of XYZ company, and your call is very important to us.
Because you are a valued customer of XYZ company, your call is important to us.

The second version also states the logical relation (*because*), and drops the empty intensifier “very”.
Principle: the grammar must say exactly what you mean. In mathematics, a sentence that says something other than what you meant is simply a wrong statement.

19.2 Choose what to say to convey the message and its spirit

Short but lifeless

The conjecture of Gause (1930) is false.

Tells the story

The lemmas of Euler (1766) and the example of Abel (1827) led Gause to conjecture that all semistable curves are modular. The conjecture was widely believed, and more than fifty papers were written by Jacobi, Dirichlet, and Galois in support of it. To everyone’s surprise and dismay, a counterexample was produced by Frobenius in 1902. This counterexample opened many doors.

(The names and dates are deliberately fictional; notice also the slide’s own typos: “beleived”, “everyones’s”, “counter example”, “Galios”.) **Principle:** you must choose what to include to convey most effectively a given message *and its spirit*. The second version tells the reader why the result matters: history, belief, surprise, consequences. Context and motivation are part of the message.

19.3 Minimise the number of notations

Bad

Let X be a compact metric subspace of the space Y . If f is continuous, $\mathbb{R}$ -valued function on the space then it assumes both a maximum and a minimum value.

What is wrong? X and Y are defined but never used again; Y is irrelevant. “The space” is ambiguous (X or Y ?). “It” is ambiguous. The article “a” is missing before “continuous”.

Good

Let X be a compact metric space. If f is a continuous, real-valued function on X then f assumes both a maximum and a minimum value.

Better

A continuous, real-valued function on a compact metric space assumes both a maximum and a minimum value.

Principle: introduce a symbol only if you will use it. Every symbol is a burden on the reader’s memory.

19.4 Prefer words to heavy symbolism

Bad: difficult to read

$$\forall x \exists y, x \geq 0 \Rightarrow y^2 = x$$

Better

Every nonnegative real number has a square root.

Logical symbols belong in logic courses, not in prose. The sentence version is shorter to read, has no ambiguity about quantifier scope, and names the concept (square root).

19.5 Be precise; avoid ambiguity

Ambiguous

All continuous functions have a maximum.

This can be read as “all continuous functions *share the same* maximum”.

Better

Every continuous function has a maximum. (More precisely: *Each* continuous function has a maximum.)

“Every” and “each” distribute over individual objects; “all” can suggest a collective property. (Mathematically the statement also needs “on a compact set”, which shows how precision matters.)

Chapter 20

General Guidelines for Technical Writing

20.1 Rules and practices of writing

1. **Be careful about the language:** grammar, sentence formation, spelling, punctuation.
2. **Each paragraph represents one specific idea.** If a paragraph has two ideas, split it.
3. **Smooth transitions** from one paragraph to the next and from one sentence to the next. Use linking words (*therefore, however, in contrast, as a result*) and let each sentence start from what the previous one ended with.
4. **Write short and simple sentences.**
5. **The opening paragraph of a section should be its best paragraph, and the opening sentence of a paragraph its best sentence:** readers skim first sentences.
6. **Every statement should be precise and correct.** “The problem stated above is difficult.” Difficult for whom? NP-complete? Believed by you? Believed by others? Proved by someone? Say exactly which.
7. **Statements should be logical; avoid sentences of the form “An x is y ”.**
 - Bad: An important method for internal sorting is quicksort.
 - Good: Quicksort is an important method for internal sorting, because ...

Put the *subject* (quicksort) first, and justify the claim.

8. **Vary sentence structure and word choice to avoid monotony, but use parallelism for parallel concepts.**
 - Bad: Formerly, science was taught by the textbook method, while now the laboratory method is employed.
 - Good: Formerly, science was taught by the textbook method; now it is taught by the laboratory method.

Two parallel ideas should have the same grammatical shape so the contrast (textbook vs laboratory) stands out.

9. **Do not omit “that” when it helps the reader parse the sentence.**
 - Bad: Assume G is a graph.

- Good: Assume that G is a graph.
10. **Sentences have a rhythm.** Read what you have written aloud and change the wording if it does not flow.
 11. **Active voice is preferred in computer science.**
 - Bad: The following result can now be proved.
 - Good: We can now prove the following theorem.
 12. **“I” or “we”?** Always use “we”, even as a single author. (“We” means the author and the reader together.)

20.2 Important points for mathematical writing

1. **Separate symbols in different formulas by words.** Bad: Consider S_q , $q < p$. Good: Consider S_q , where $q < p$.
2. **Do not start a sentence with a symbol.** Bad: G has n vertices. Good: The graph G has n vertices. (A capital letter cannot mark the sentence start if it is a symbol, and “ n vertices. $G \dots$ ” reads like a formula.)
3. **Do not use $\forall$, $\exists$, $\in$ in prose;** write “for all”, “there exists”, “in”.
4. **Define symbols before use.**
 - Bad: Algorithm XYZ finds a drawing of G in $O(n + m)$ time, where n and m are the numbers of vertices and edges, respectively.
 - Good: Let G be a graph of n vertices and m edges. Then Algorithm XYZ finds a drawing of G in $O(n + m)$ time.
5. **Do not often use quotations in mathematics papers.** When you do, punctuation follows American style: **commas and periods go inside** the quotation marks; **colons and semicolons go outside**.
 - Bad: As Methuselah used to say, “When the going gets tough, the tough get going”.
 - Good: As Methuselah used to say, “When the going gets tough, the tough get going.”
6. **The statement just before a theorem, algorithm, etc. should be a complete sentence or end with a colon.**
 - Bad: We now have the following **Theorem.** $H(x)$ is continuous.
 - Good: We can now prove the following result. **Theorem.** The function $H(x)$ defined in (5) is continuous.
7. **A theorem statement should be self-contained**, not depending on assumptions in the previous text. Notice the good version restates “the function $H(x)$ defined in (5)”. Readers often read only the theorems.
8. **Capitalise names of specific numbered items:** Theorem 1, Lemma 2, Algorithm 3, Table 5, Figure 4, Section 3.
 - Wrong: By lemma 3, we have ... Correct: By Lemma 3, we have ...
 - Wrong: We now have the following Lemma. Correct: We now have the following lemma.

- Wrong: ...illustrated in figure 5(a). Correct: ...illustrated in Figure 5(a).
- Wrong: In section 3 we deal with ... Correct: In Section 3 we deal with ...

Rule: capitalise when it is a *name* with a number; lower case when it is a common noun (“the following lemma”).

9. Spell out small numbers used as adjectives; use digits when the number is a name or value.

- Wrong: There are 5 vertices on the outer face. Correct: There are five vertices on the outer face.
- Wrong: The count was increased by two. Correct: The count was increased by 2. (2 is a value.)
- Wrong: The graph has eighty embeddings. Correct: The graph has 80 embeddings. (large number)

Common convention: spell out zero to nine (or ten) as adjectives; use digits for larger numbers, values, measurements and anything with a unit.

- 10. Display important formulas on a line by themselves**, and number the important ones even if you do not reference them all (others may want to cite them).
- 11. Do not overuse commas.** Bad: “I went to the store, to buy some potatoes.” The slides even flag “In this paper, we give a linear-time algorithm” as over-punctuated; a short introductory phrase needs no comma.
- 12. No contractions in formal writing:** “do not”, not “don’t”; “I am”, not “I’m”; “it is”, not “it’s”.

20.3 Fact, Lemma, Theorem, Corollary

All of these are *propositions*: statements that are true or false.

Type	Meaning	Proof?
Fact	obviously true or well known	usually no
Lemma	a helper result, used to prove other propositions	yes
Theorem	a main result of the paper	yes
Corollary	follows immediately from a theorem or lemma	short / immediate
Proposition	a result of moderate importance	yes
Definition	introduces a term (not true/false)	no
Conjecture	believed true but not proved	–

Exam focus

In the safelabeling write-up, “Theorem 2” on trees follows immediately from Theorem 1 (a tree is bipartite), so it should really be a **Corollary**. Also, it defines m and $\Delta(T)$ but never uses them: remove them (minimise notation).

20.4 How to write a theorem and a proof

- 1. State before you prove.** Introduce the result in a complete sentence: “We now prove that every tree admits a k -safe labeling of span $n + k - 1$.”

2. **Make the statement self-contained:** include every hypothesis (“Let T be a tree of n vertices. Then ...”).
3. **Tell the reader the strategy first:** “We prove the theorem by induction on n ” or “The proof is by contradiction.”
4. **Use lemmas** to break a long proof into readable pieces.
5. **Justify each step** with a reason (“by Lemma 1”, “since G has no odd cycle”).
6. **Mark the end** with $\square$ (the `proof` environment does this).
7. **Write proofs in sentences**, not chains of symbols and arrows.

```

\begin{theorem}\label{thm:bip}
Let  $G$  be a bipartite graph with  $n$  vertices
and bipartition  $V = P \cup Q$ . Then  $G$ 
admits a  $k$ -safe labeling with span  $n+k-1$ .
\end{theorem}
\begin{proof}
Label the vertices of  $P$  with  $1, 2, \dots, |P|$ 
and the vertices of  $Q$  with
 $|P|+k, \dots, |P|+k+|Q|-1$ . Every edge joins a
vertex of  $P$  to a vertex of  $Q$ , so the two labels
of an edge differ by at least  $k$ . The largest
label is  $n+k-1$ , so the span is  $n+k-1$ .
\end{proof}
\begin{corollary}
Every tree with  $n$  vertices admits a  $k$ -safe
labeling with span  $n+k-1$ .
\end{corollary}
\begin{proof}
A tree has no cycle, hence no odd cycle, so it is
bipartite; apply Theorem~\ref{thm:bip}.
\end{proof}

```

Theorem 20.1. *Let G be a bipartite graph with n vertices and bipartition $V = P \cup Q$. Then G admits a k -safe labeling with span $n + k - 1$.*

Proof. Label the vertices of P with $1, 2, \dots, |P|$ and the vertices of Q with $|P|+k, \dots, |P|+k+|Q|-1$. Every edge joins a vertex of P to a vertex of Q , so the two labels of an edge differ by at least k . The largest label is $n + k - 1$, so the span is $n + k - 1$. $\square$

Corollary 20.2. *Every tree with n vertices admits a k -safe labeling with span $n + k - 1$.*

Proof. A tree has no cycle, hence no odd cycle, so it is bipartite; apply Theorem 20.1. $\square$

20.5 Worked example: correcting the safelabeling write-up (Quiz Q2)

The quiz gives a paper that purposefully violates the principles. Here are the errors and fixes, grouped by principle, so you can spot them in any new text.

Line	Problem	Correction
2	“Dr.” in author name	Umme Habiba

Line	Problem	Correction
3–5	affiliation style	put affiliation under the name without title “Associate Professor” (or in a footnote)
7	$G' = (V', E')$ is a subgraph if $V = V'$, $E' \subseteq E$: notation defined, never used	remove it (minimise notation); also it defines a <i>spanning</i> subgraph
7	“I wish to label”	“We wish to label” (always “we”)
9	“We will call” after “I wish”	consistent “we”
13	“I will give”, space in “ k -safe”	“we give”; “ k -safe”
13	abstract promises upper bounds on trees and bipartite graphs; keywords “Minimum, Small Span”	keywords should be meaningful terms: “safe labeling, graph labeling, span, bipartite graphs, trees”
14	citations [4, 5] in the abstract	avoid citations in an abstract
19	“graph[1]”	“graph [1]” (space, tie)
19	“integer .”	remove space before period
21	“ k -safe labeling of g ”	“of G ” (consistent symbol)
22	“range from the largest to the smallest”	“from the smallest to the largest” (consistent with abstract)
23	starts sentence with symbol “ $I_l - I_s + 1$ is the span” and uses I_s, I_l before defining them	“Let I_s and I_l be the smallest and largest integers. Then the span is $I_l - I_s + 1$.”
25	space before period “span .”	remove
26	“In figure 1”	“In Figure 1”
26	“graph of 6 vertices”	“six vertices”
27	“thirty, 13 and 10” inconsistent	“30, 13 and 10”
27	“respectively .”	remove space
28	starts sentence with symbol “ K_n needs...”; claim without citation or proof	“The complete graph K_n needs...” and justify
30–31	“Thus one may expect... narrow spans” vague	state precisely what is shown
33	“simple graph .”	remove space
39	“then G is called a tree T ”	a symbol cannot be both G and T : “is called a tree”
44	“The following Lemmas”	“the following lemmas” (common noun)
46	“vertices of g ”	“of G ”
48	“in lemma 2,”	“in Lemma 2” and no comma before “leads”
53	“It is interesting to observe that”	weak filler; delete: “The span $n + k - 1$ in Theorem 1 is optimal...”
53	“theorem 1”, “trivial span”	“Theorem 1”; “trivial” is misleading
56	“following Theorem”	“following theorem”
57–58	m and $\Delta(T)$ defined but unused; result is immediate	remove; make it a Corollary
Proof	“by lemma 1 in section 2”, “by theorem 1”	“by Lemma 1 in Section 2”, “by Theorem 1”
60	“So this is a great paper.”	unprofessional, unsupported; summarise contributions and future work
62–73	inconsistent reference style: name order (J. A. Gallian vs Gross J. L.), quotes vs none, “pp. ,” empty, year duplicated, “(2001)” in parentheses, lowercase “ofdma”, “WLANS”	one consistent style (BibTeX); complete page numbers; protect acronyms: OFDMA, WLANS

20.6 Referee reports and book reviews

The syllabus includes writing reviews. A referee report typically has: (1) a short **summary** of the paper in your own words, showing you understood it; (2) an assessment of **novelty, correctness, significance and clarity**; (3) **major comments** (errors in proofs, missing comparisons, unsupported claims); (4) **minor comments** (typos, notation, figures) with page/line numbers; (5) a **recommendation**: accept, minor revision, major revision, or reject. Be specific, polite and constructive; criticise the work, not the authors. A book review summarises the book’s aim, audience and content, then evaluates coverage, accuracy, clarity and usefulness.

20.7 Sources used in the slides

D. E. Knuth, T. Larrabee and P. M. Roberts, *Mathematical Writing*, MAA Notes 14, 1989. S. G. Krantz, *A Primer of Mathematical Writing*, AMS, 1997. R. Andonie and I. Dzitac, “How to write a good paper in computer science and how will it be measured by ISI Web of Knowledge”, *Int. J. of Computers, Communications & Control*, 2010. U. Khedker, *How to Write a Good Paper?* (IIT Bombay slides).

Part V

Research: Reading and Writing Papers

Chapter 21

How to Read a Research Paper

21.1 Why and what you read

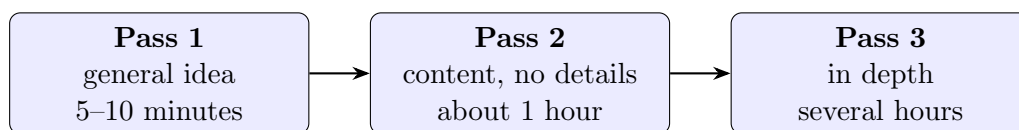
You read papers to **review** them (for a conference, journal or class: read fully and critically), to **keep up to date** in your field (skim), or to do a **literature review** of a new field (read many, find connections). Before investing time, filter on:

Topic	Is it the right paper? Experimental or theoretical?
Venue	Reputed venue? Journal or conference?
Quality	Citation count? Impact?
Authors	Reputed in the area? Recent work?

21.2 IMRaD

Most scientific papers follow **I**ntroduction, **M**ethods, **R**esults and **D**iscussion: the problem, how it was studied, what was found, what it means. Knowing the structure tells you where to look.

21.3 The three-pass approach



Pass 1 (bird’s-eye view). Read title, abstract, introduction, section headings and conclusion; glance at references. Then answer the **five Cs**:

1. **Category:** measurement paper, analysis of an existing system, new algorithm or prototype?
2. **Context:** which papers and theories does it build on?
3. **Correctness:** do the assumptions appear valid?
4. **Contributions:** what are the main contributions?
5. **Clarity:** is it well written?

Pass 2 (grasp the content). Read carefully but skip proofs. Study figures: labelled axes? error bars? statistically meaningful? Mark unread references. You should now be able to summarise

the paper, with evidence, to someone else. If you still do not understand it (new subject, jargon, forward references, poor writing), either set it aside, return after background reading, or persevere to pass 3.

Pass 3 (virtually re-implement). Make the same assumptions as the authors and re-create the work; compare. Afterwards you can reconstruct the paper from memory, identify strong and weak points, pinpoint implicit assumptions and missing citations, and see issues with experimental or analytical techniques. This is where your own research ideas come from.

21.4 Doing a literature survey

1. Use **Google Scholar** with good keywords; find 3–5 recent, highly cited papers; do one pass on each.
2. Look for a **survey paper**; if none, the related-work sections of your 3–5 papers may point to one.
3. Find the **key people**: authors and citations that appear repeatedly; visit their web pages.
4. Find the **top venues**: where the key people publish.
5. **Download and read** recent proceedings and archives using the three-pass method.

21.5 Read critically, creatively, and take notes

Critically: Is the problem relevant? Is it new knowledge? Is the design appropriate and the protocol right? Is there bias? Are conclusions justified? **Creatively:** What are the good ideas? Can they be extended or applied elsewhere? Which results are influential? What improvements are possible? **Take notes** (electronic or written), because you will forget.

21.6 Summarising

A summary states the problem, approach, main result and significance in your own words, keeps the author's meaning, and adds no opinion (critique goes separately). **Automatic summarisation** is *extractive* (selects key sentences) or *abstractive* (generates new sentences, e.g. language models); always check it against the paper.

Chapter 22

How to Write a Research Paper

22.1 Before writing

- **What do I have to say?** State the question and its answer in one simple sentence. Too much in one paper makes it diffuse.
- **Salami science** (salami slicing): splitting one study into several papers to inflate publication count (“publish or perish”: quantity over quality). It wastes journal space and reviewers’ and readers’ time. Splitting is justified only for genuinely different audiences or an overly long manuscript.
- **Is it worth writing?** Check the literature. A paper takes 20–30 drafts.
- **Format:** original research article; short communication or letter; review (like a focused mini-book). **Journal** (longer, several review rounds, no fixed deadline) or **conference** (deadline, one review round, oral presentation; the main venue type in CS).
- **Venue:** check scope, allowed format, audience match and impact; take advice from a mentor. A wrong choice costs time or leads to rejection. Parameters: impact factor, SCImago rank (Q1–Q4), editorial board, publisher (avoid predatory publishers, e.g. via Beall’s list), indexing (Scopus, Web of Science).

22.2 The paper as a critical argument

Problem statement posing a question → presentation of evidence → assessment of validity (strengths, weaknesses, other evidence) → conclusions.

22.3 Literature search: high sensitivity, low specificity

Be thorough: *high sensitivity* (few false negatives: you miss few relevant papers) at the cost of *low specificity* (many false positives you then filter). Missing a key paper is worse than reading irrelevant ones.

22.4 Title, authorship and planning

Start with a tentative title. Fix authorship and author order **early** to avoid hurt feelings; conventions: alphabetical, corresponding author, senior author last, main contributor first. Acknowledge

funding and helpers. Create an **outline first**, plan multiple drafts (dated filenames or version control such as Git), make **tables and figures early** (they prompt analysis), and set deadlines for yourself and co-authors.

22.5 Section by section

Introduction. Draw the reader in; identify the gap; end with the question or hypothesis. In CS, answer the **five questions**:

1. What is the problem?
2. Why is it interesting and important?
3. Why is it hard? (Why do naive approaches fail?)
4. Why hasn't it been solved before? (What is wrong with previous solutions? How does mine differ?)
5. What are the key components of my approach and results?

Methods. Subheadings; clear enough for a non-specialist; enough detail to *replicate*; past tense.

Results. Organise around tables and figures; past tense; facts only, no interpretation; figures for trends, tables for exact values; every table and figure self-contained (understandable from its caption); extras in supplementary material.

Discussion. First answer the question and interpret the main results; then evidence for and against, implications (be conservative), other findings, limitations; finally concluding remarks and future directions.

Abstract. A mini-paper: 1–2 sentences of introduction, methods (often longest), results (no tables or figures), one concluding statement. It needs many drafts.

22.6 Writing tips

- **Tense:** others' published work in the present ("Smith shows"); your own study, methods and results in the past ("we measured").
- **Voice:** some fields demand passive; CS prefers active ("We ran 34 tests"); a mix is common.
- **Brevity:** every word must count.
- **Top-down check:** for each section, paragraph, sentence and word ask whether it serves the level above; if not, rewrite or delete.
- **Proofread:** consistent terms; abstract numbers match text and tables; in-text citations match references; grammar.

Chapter 23

Seven Suggestions for Writing a CS Paper

From Simon Peyton Jones (Microsoft Research Cambridge).

1. **Don't wait: write.** Model 2 (idea → write paper → do research) beats Model 1 (idea → research → write). Writing forces clarity, exposes what you don't understand, and opens dialogue with others. Writing is a primary mechanism for *doing* research, not just reporting it.
2. **Identify your key idea.** Convey one useful, re-usable idea (“infect the reader's mind like a virus”). Don't be intimidated: write about any idea, however small it seems. A paper has one clear “ping”; say it explicitly: “The main idea of this paper is ...”. Many ideas? Write many papers.
3. **Tell a story.** As at a whiteboard: here is a problem; it is interesting; it is unsolved; here is my idea; my idea works (details, data); here is how it compares with others. Structure: title (1000 readers), abstract (4 sentences, 100), introduction (1 page, 100), problem (1 page, 10), idea (2 pages, 10), details (5 pages, 3), related work (1–2 pages, 10), conclusions (0.5 page).
4. **Nail your contributions to the mast.** The one-page introduction describes the problem (with an example) and states the contributions as a **bulleted list**, written first. Contributions must be **refutable**: “We prove that the type system is sound (Section 4)”, not “We study its properties”. Each claim needs evidence (theorems, measurements, comparisons, case studies), forward-referenced. No “The rest of this paper is structured as follows” paragraph; use forward references in the narrative.
5. **Related work: later.** After your idea, because the reader cannot understand compressed trade-offs before knowing the problem, and it stands between the reader and your idea. Credit is not like money: be generous to competitors, acknowledge helpers and your own weaknesses.
6. **Put your readers first.** Explain intuition first, with examples, then the general case; do not describe your personal journey of discovery; avoid dense formalism that makes readers feel stupid.
7. **Listen to your readers.** Use friendly “guinea pigs” (experts and non-experts; each can read it fresh only once) and ask where they got lost. Send the draft to competitors to check you describe their work fairly. Treat every review like gold dust; fix the paper rather than arguing, and thank reviewers.

23.1 Language and style

Meet deadlines and page limits (never shrink margins or fonts); put extra proofs or data in an appendix; always spell-check; give strong visual structure (sections, bullets, italics, laid-out code, pictures). Use active, simple language:

Avoid	Prefer
It can be seen that ...	We can see that ...
34 tests were run	We ran 34 tests
The object under study was displaced horizontally	The ball moved sideways
On an annual basis	Yearly
Endeavour to ascertain	Find out

Part VI

Exam Preparation

Chapter 24

Exam Traps and Quick Answers

Question	Answer and why
Bold with a declaration	<code>\bfseries bold</code> , not <code>\bfseries{bold}</code>
Obsolete <code>\bf</code> , <code>\it</code> , <code>\em</code>	declarations; they affect everything after, until the group ends
Minimum document	<code>\documentclass,</code> <code>\begin{document},</code> <code>\end{document}</code>
Class for research papers	<code>article</code>
Class with chapters	<code>report</code> , <code>book</code>
Not a list environment	<code>record</code> (lists: <code>itemize</code> , <code>enumerate</code> , <code>description</code>)
Order of float letters	irrelevant: always tried as h, t, b, p
[H]	exact placement, needs package <code>float</code>
Column spec goes in	braces <code>{ lcccr }</code> , not brackets
<code>\begin{thebibliography}{2}</code>	widest label one digit: up to 9 items
<code>\begin{thebibliography}{99}</code>	up to 99 items
Reference numbering	by list order, by mention (<code>unsrt</code>), alphabetical (<code>plain</code>); never random
Label placement in a figure	after <code>\caption</code>
Caption position	tables above, figures below
<code>\$\$..\$\$</code> vs <code>\[..\]</code>	both display unnumbered; prefer <code>\[..\]</code>
<code>equation</code> vs <code>equation*</code>	numbered vs unnumbered
Invisible content of a given size	<code>\phantom</code> , <code>\hphantom</code> , <code>\vphantom</code>
Taller table rows	<code>\renewcommand{\arraystretch}{2}</code>
Frame vs slide	frame = source block; slide = PDF page; overlays make many slides per frame
<code>\only</code> vs <code>\uncover</code>	<code>\only</code> removes content (no space); <code>\uncover</code> reserves space
Verbatim in beamer	frame option <code>[fragile]</code>
TikZ default unit	1 cm
TikZ statement terminator	semicolon
<code>(30:2)</code>	polar: 2 cm at 30°
<code>++(1,0)</code> vs <code>+(1,0)</code>	both relative; <code>++</code> updates the current point
<code>right=of a</code>	needs <code>positioning</code> library
Vector vs raster	vector for diagrams/plots (scalable); raster for photos
Pie vs bar chart	pie: parts of a whole; bar: compare quantities
Presentation practice	few slides, enough time for each
Visible slides	large fonts, high contrast
Why LaTeX	maths typesetting; automatic numbering/references; consistent professional formatting (also BibTeX, version control)
“I” or “we”	always “we”
“lemma 3”	“Lemma 3”; but “the following lemma”
“5 vertices”	“five vertices”; but “increased by 2”, “80 embeddings”

Question	Answer and why
Punctuation and quotes	commas and periods inside; colons and semicolons outside
Sentence starting with a symbol	rewrite: “The graph G has...”
Five Cs (Pass 1)	category, context, correctness, contributions, clarity
Five Qs (Introduction)	problem, why important, why hard, why unsolved, key components
Salami science	splitting one study into many papers
Literature search	high sensitivity, low specificity

Chapter 25

Practice Problems with Solutions

25.1 Problems

1. Write the code for a table with four columns (first left-aligned, others right-aligned) using `booktabs`, with a caption “Running times” above it and label `tab:rt`.
2. What is printed by `\textbf{A \textit{B} C}` and by `{\itshape A \textbf{B}} C`?
3. Write an `align` environment of three lines where only the last line is numbered.
4. Write TikZ code for a triangle with vertices $A(0,0)$, $B(3,0)$, $C(1,2)$, labelled at the vertices, with the side AB labelled c below its midpoint.
5. Draw a binary tree with root 5, children 3 and 8, and 8 having a right child 9.
6. Write a beamer frame where three items appear one at a time and the second item turns red on slide 3.
7. Explain the output of `\foreach \x in {1,3,...,9} \fill (\x,0) circle (2pt);`.
8. Correct this sentence: “In figure 3, G has 7 vertices, it can be seen that the graph is bipartite.”
9. Why should a theorem statement be self-contained? Rewrite “Theorem. It is planar.”
10. What goes wrong with `\begin{figure}\label{f}\caption{X}\end{figure}` and `Figure \ref{f}`?

25.2 Solutions

1.

```
\begin{table}[htbp]
\centering
\caption{Running times}\label{tab:rt}
\begin{tabular}{@{}lrrr@{}}
\toprule
Algorithm &  $n=10^3$  &  $n=10^4$  &  $n=10^5$  \\
\midrule
BFS & 0.2 & 1.9 & 20.4 \\
\bottomrule
\end{tabular}
\end{table}
```

2. First: **A B C** (all bold, B bold italic). Second: *A* ***B*** C (A italic, B bold italic, C normal: the group ends before C).

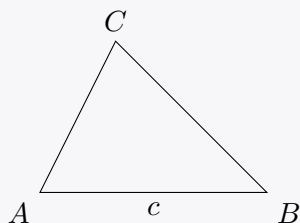
3.

```
\begin{align}
x &= a + b \nonumber \\
&= c + d \nonumber \\
&= e
\end{align}
```

$$\begin{aligned} x &= a + b \\ &= c + d \\ &= e \end{aligned} \tag{25.1}$$

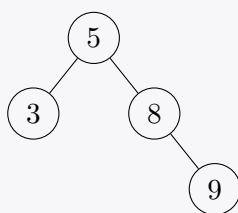
4.

```
\begin{tikzpicture}
\coordinate[label=below left:$A$] (A) at (0,0);
\coordinate[label=below right:$B$] (B) at (3,0);
\coordinate[label=above:$C$] (C) at (1,2);
\draw (A) -- node[below]{$c$} (B)
-- (C) -- cycle;
\end{tikzpicture}
```



5.

```
\begin{tikzpicture}[level distance=10mm,
sibling distance=16mm,
every node/.style={circle,draw}]
\node {5}
child {node {3}}
child {node {8}}
child[missing]
child {node {9}}};
\end{tikzpicture}
```



6.

```

\begin{frame}{Steps}
  \begin{itemize}
    \item<1-> First
    \item<2-> \alert<3>{Second}
    \item<3-> Third
  \end{itemize}
\end{frame}

```

7. The list $\{1, 3, \dots, 9\}$ infers step 2, so it draws five filled dots at $x = 1, 3, 5, 7, 9$ cm on the x -axis.
8. “Figure 3 shows a graph G with seven vertices. We can see that G is bipartite.” (Capital F, no sentence beginning with a symbol, “seven”, no comma splice, active voice.)
9. Readers often read only theorem statements; each must stand alone. “**Theorem.** The graph G constructed in Section 3 is planar.”
10. The `\label` comes before `\caption`, so it refers to the current section, not the figure: “Figure 2” may print the section number. Put `\label` after `\caption`. Also use `Figure~\ref{f}` so the number does not break onto a new line.

Chapter 26

One-Page Cheat Sheet

Structure

```
\documentclass[12pt]{article}
\usepackage{amsmath,graphicx}
\title{} \author{} \date{}
\maketitle \tableofcontents
\section{} \subsection{}
\label{sec:x} \ref{sec:x} \cref{}
```

Text

```
\textbf{} \textit{} \emph{} \texttt{}
{\bfseries } {\itshape } {\large }
\footnote{} \url{} \href{}{}
~ (tie) \\ (break) \par
\hspace{} \vspace{} \phantom{}
```

Lists

```
itemize, enumerate, description
\item \item[label]
```

Maths

```
$..$ \[. .] equation align
x^{2} x_{i} \frac{}{} \sqrt{}
\sum_{i=1}^n \int_a^b \lim
\left( \right) \text{}
\mathbb{R} \mathcal{O} \mathbf{v}
pmatrix bmatrix cases
\newtheorem{thm}{Theorem}
```

Tables

```
\begin{tabular}{|l|c|r|}
a & b & c \\ \hline
\multicolumn{2}{c}{} \multirow{2}{*}{}
\toprule \midrule \bottomrule
\renewcommand{\arraystretch}{1.5}
```

Floats

```
\begin{figure}[htbp] \centering
\includegraphics[width=.8\linewidth]{}
\caption{} \label{fig:x}
```

Bibliography

```
\cite{key}
\bibliographystyle{plain}
\bibliography{refs}
\begin{thebibliography}{99}
\bibitem{key} ...
```

TikZ

```
\draw[->,thick] (0,0) -- (1,1);
\draw (0,0) circle (1);
\draw (0,0) rectangle (2,1);
\node[draw,circle] (a) at (0,0) {A};
\node[right=of a] (b) {B};
\draw (a) -- node[above]{x} (b);
\foreach \i in {1,...,5} {...}
\fill[red] (1,1) circle (2pt);
name/.style={...}
```

Beamer

```
\documentclass{beamer}
\usetheme{Madrid}
\begin{frame}{Title} ... \end{frame}
\titlepage \pause
\item<2-> \only<2>{} \uncover<2>{}
\alert<3>{} block alertblock
columns \column{.5\textwidth}
[fragile] [allowframebreaks]
```

Writing rules

we, not I; active voice; define symbols first; no symbol at sentence start; words not $\forall\exists$; “Lemma 3”, “Figure 5”; five (adjective) vs 80 (number); no contractions; self-contained theorems; one idea per paragraph; best sentence first.

CSE 300 Comprehensive Practice Set
Bangladesh University of Engineering and Technology
L-3/T-1 CSE 300: Technical Writing and Presentation

Time: 90 minutes

Marks: 100

Student Name: Kazi Asef Kabin

Student No: 2305084

Section A: Multiple Choice Questions

(15×1=15)

Write the letter of the most appropriate answer, e.g. (a): (C).

- ✓ (a) Which command produces bold text limited to one word?
(A) ~~\bfseries {word}~~ (C) \bf {word}
✓ (B) {\bfseries word} (D) \bold {word}
- ✓ (b) Which package provides \toprule, \midrule and \bottomrule?
(A) array (C) ~~booktabs~~
(B) tabularx (D) multirow
- ✓ (c) Which float specifier places a figure exactly at its position in the source?
(A) [h] (C) [t]
(B) [h!] (D) ~~[H]~~
- ✓ (d) In TikZ, what does (45:2) denote?
(A) The point (45,2) (C) 45 cm at 2°
✓ (B) 2 cm at an angle of 45° (D) A relative move
- ✓ (e) Which TikZ library is needed for below=of A?
(A) calc (C) ~~positioning~~
(B) shapes (D) arrows.meta
- ✓ (f) In beamer, which command hides content but still reserves its space?
(A) \only (C) \alert
✓ (B) \uncover (D) \temporal
- ✓ (g) Which frame option is required to use \verb inside a beamer frame?
(A) plain (C) ~~fragile~~
(B) shrink (D) allowframebreaks
- ✓ (h) Which BibTeX style numbers references in order of citation?
(A) plain (C) ~~abbrv~~
(B) alpha (D) ~~unsrt~~
- ✓ (i) Which BibTeX entry type fits a paper in a conference proceedings?
✓ (A) @article (C) @book
• @inproceedings (D) @misc
- ✓ (j) Which statement is *not* one of the “five Cs” of the first reading pass?
(A) Category (C) ~~Complexity~~
(B) Context (D) Clarity
- ✓ (k) Reporting one study in several papers to increase publication count is called
(A) Plagiarism (C) Peer review
✓ (B) Salami science (D) Self-citation
- ✓ (l) According to Peyton Jones, where should related work be placed?
(A) Right after the abstract (C) ~~After your main idea~~
(B) In the introduction (D) It should be omitted

- (m) Which sentence follows the CSE 300 writing rules?
- (A) G has n vertices. (C) G has n vertices.
 (B) The graph G has n vertices. (D) $\forall v \in G, v$ is a vertex.
- (n) A proposition that follows immediately from a theorem is a
- (A) Lemma (C) Corollary
 (B) Fact (D) Conjecture
- (o) What will `\foreach \x in {0,2,...,8}` iterate over?
- (A) 0, 2, 8 (C) 0, 2, 4, 6, 8
 (B) 0, 1, 2, ..., 8 (D) 2, 4, 6, 8

Section B: True or False

(10×1=10)

Write T or F. Correct each false statement in one line.

- (a) The `report` class supports the `\chapter` command. **T**
- (b) `[htbp]` and `[bpht]` produce different placements. **F** → always $h \rightarrow t \rightarrow b \rightarrow p$
- (c) `\label` placed before `\caption` in a figure refers to the figure number. **T** → **F** → put label after `\caption`
- (d) Text inside `\[ ... \]` is numbered automatically. **T** → **F** → unnumbered
- (e) In TikZ, every drawing command must end with a semicolon. **T**
- (f) In TikZ, `scale=2` also doubles the size of node text by default. **F** → scales only coordinate/shape
- (g) One beamer frame can produce several slides. **T**
- (h) In computer science writing, the passive voice is preferred. **F**
- (i) "There are five vertices on the outer face" is correctly written. **F** → **T** → $<10 \rightarrow$ in words, $>10 \rightarrow$ numbers
- (j) High sensitivity in a literature search means you rarely miss relevant papers. **T**

Section C: Fill in the Blanks

(10×1=10)

- (a) The argument `{99}` in `\begin{thebibliography}{99}` specifies the widest label.
- (b) To double the row height of a table, write `\renewcommand{\arraystretch}{2}`.
- (c) The command `\phantom` creates an invisible box of the same size as its argument.
- (d) In TikZ, `++(1,0)` moves relative to the last point and updates the current point.
- (e) The beamer command `\titlepage` shows the title page.
- (f) Table captions are conventionally placed above the table, figure captions below the figure.
- (g) The standard structure of a scientific paper is abbreviated IMRaD.
- (h) In pass 3 of reading a paper, you virtually re-implement the work.
- (i) Commas and periods go inside quotation marks; colons and semicolons go outside.
- (j) The TikZ path operation `-- cycle` closes a path back to its starting point.

Section D: Matching

(8×1=8)

Match each item in Column I with the best item in Column II (e.g. 1-e).

Column I	Column II
1 <code>\emph {}</code>	a spans several columns of a table
2 <code>\multicolumn</code>	b reference to an equation with parentheses
3 <code>\eqref</code>	c semantic emphasis
4 <code>\pause</code>	d draws a gradient fill in TikZ
5 <code>\shade</code>	e splits a frame into several slides
6 <code>\cite</code>	f surrounds nodes with a box
7 <code>fit</code>	g number of a bibliography entry
8 <code>\tableofcontents</code>	h list of sections

Section E: Predict the Output

(4×3=12)

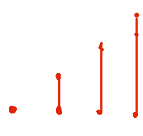
Describe or sketch the output of each snippet. → **bold**

(a) `{\itshape Graph \textbf{labeling}} is \emph{fun \emph{really}}}`.

(b) `\begin{align}`

$$\begin{array}{l} a \&= b + c \\ \&= d \end{array}$$

`\end{align}`
italic $a = b + c$ $= d$ ① *normal*

(c) `\begin{tikzpicture}`
`\foreach \i in {0,...,3}`
`\draw (\i,0) -- (\i,\i);`
`\end{tikzpicture}`


(d) `\begin{frame}{Demo}`
`\begin{itemize}`
`\item<1-> A`
`\item<2-> B`
`\item<3-> \alert<3>{C}`
`\end{itemize}`
`\end{frame}`
slide-1: *A* *slide-2:* *A B* *slide-3:* *A B* C

How many slides are produced, and what is on each?

Section F: Find and Fix the Errors

Each snippet has at least two errors. Identify and correct them.

(a) `\begin{table}[htbp]`
`\begin{tabular}{lcr}` → *{lcr}*
`Name & Age & Score \&` → *\&*
`Rafi & 22 & 90`
`\end{tabular}`
`\label{tab:x}\caption{Scores}` → *label must come after caption*
`\end{table}`
\hline

(b) `\begin{tikzpicture}`
`\node[draw] (a) at (0,0) {A};`
`\node[draw, right=of a] (b) {B};`
`\draw[->] (a) -- (b);`
`\end{tikzpicture}`
for that use tikzlibrary {positioning} needed ①

(c) `\begin{frame}{Code}`
`\begin{verbatim}`
`int x = 0;`
`\end{verbatim}`
`\section{Results}`
`\end{frame}`
[fragile] for verbatim

* In figure → *label first* (3×4=12)
 * In table → *caption first*

Section G: Correct the Writing

(5×2=10)

Rewrite each sentence following the CSE 300 writing rules.

(a) In section 4.6, prove the following Theorem.

(b) T is a tree with 6 vertices, it has 5 edges.

(c) An efficient algorithm for sorting is merge sort.

(d) The problem stated above is very difficult and it's not easy to solve.

(e) Algorithm ABC runs in $O(n^2)$ time where n is the number of vertices of G .

Section H: Short Questions

(a) Differentiate between \only and \uncover with an example.

(b) What is the difference between the tabular and table environments?

(c) Give two reasons to prefer vector graphics over raster graphics in a paper.

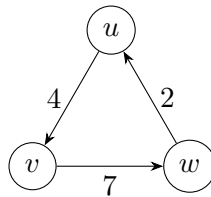
(d) Differentiate a lemma from a theorem.

(e) What are the five questions an introduction should answer?

(f) Why should you "nail your contributions to the mast"? What makes a contribution refutable?

Section I: Write the Code

(a) Write a TikZ picture of a directed cycle on three nodes u , v , w (circles) with edge weights 4, 7 and 2.



(b) Write a numbered equation environment for the piecewise function $f(n) = 1$ if $n = 0$, and $n \cdot f(n - 1)$ otherwise, labelled eq:fact, and a sentence referring to it.

(c) Write a beamer frame with two columns: a bulleted list of two items on the left, and a block titled "Theorem" on the right that appears only from slide 2.

Section J: Long Answer

(1×11=11)

Explain the three-pass approach to reading a research paper. For each pass, state what you read, the goal, and what you should be able to do afterwards. Then explain how you would use it to survey a new research area.

1. Pass -1: 5-10 mins → Just idea → Abstract, Conclusion

2. Pass -2: 1 hr → Read methodology & other stuffs

3. Pass -3: Several hours → in depth discussion

J. a) `\begin{tikzpicture} [>=stealth, v\style={circle,draw,minimum size=6mm}`

`\node[v] (v) at (90:1.2) {$v$};`

`\node[v] (v) at (210:1.2) {$v$};`

`\node[v] (w) at (330:1.2) {$w$};`

`\draw [->] (v) -- node[left] {4} (v);`

`\draw [->] (v) -- node[below] {7} (w);`

`\draw [->] (w) -- node[right] {2} (v);`

`\end{tikzpicture}`

b) `\begin{equation}`

`f(n) = \begin{cases}`

`1 \quad \&\text{if } n=0 \quad \backslash\backslash`

`n \cdot f(n-1) \quad \&\text{otherwise}`

`\end{cases}`

`\label{eq:fact}`

`\end{equation}`

Equation `\eqref{eq:fact}` defines the factorial.

c) `\begin{frame} Two columns`

`\begin{columns}`

`\column{0.5\textwidth}`

`\begin{itemize}`

`\item First item`

`\item Second item`

`\end{itemize}`

`\column{0.5\textwidth}`

`\uncover{2->}`

`\begin{block} Theorem`

Every Tree is bipartite.

`\end{block}`

`\end{columns}`

`\end{frame}`

CSE 300 Comprehensive Practice Set 2
Bangladesh University of Engineering and Technology
L-3/T-1 CSE 300: Technical Writing and Presentation

Time: 90 minutes

Marks: 100

Student Name: Kazi Asef Kabir

Student No: 2305084

Section A: Multiple Choice Questions

(15×1=15)

Write the letter of the most appropriate answer, e.g. (a): (C).

- (a) Which is a valid way to write a comment in LaTeX?
(A) `// comment` (C) `% comment`
(B) `# comment` (D) `<!-- comment -->`
- (b) Which command gives a non-breaking space?
(A) `\,` (C) `\\`
(B) `~` (D) `\space`
- (c) Which environment should be used for several aligned, numbered equations?
(A) `eqnarray` (C) `gather*`
(B) `align` (D) `equation`
- (d) Which command spans a cell over three rows?
(A) `\multicolumn {3}{c}{x}` (C) `\rowspan {3}{x}`
(B) `\multirow {3}{*}{x}` (D) `\cline {1-3}`
- (e) Which of these is a declaration, not a command with an argument?
(A) `\textit` (C) `\large`
(B) `\emph` (D) `\underline`
- (f) In TikZ, which draws a filled red circle of radius 3pt at (1,1)?
(A) `\draw [red] (1,1) circle (3pt);` (C) `\circle [red]{1,1}{3pt};`
(B) `\fill [red] (1,1) circle (3pt);` (D) `\node [red] (1,1) {3pt};`
- (g) What does `\draw (0,0) -| (2,1);` draw?
(A) A diagonal line (C) Vertical then horizontal
(B) Horizontal then vertical (D) A rectangle
- (h) Which TikZ option makes a label follow the angle of a line?
(A) `rotate` (C) `auto`
(B) `sloped` (D) `midway`
- (i) Which beamer command produces a red highlight?
(A) `\alert` (C) `\only`
(B) `\pause` (D) `\uncover`
- (j) The overlay specification `<-3>` means
(A) only slide 3 (C) slides 1 to 3
(B) slide 3 onwards (D) all slides except 3
- (k) Which is *not* a beamer theme?
(A) Madrid (C) Berlin
(B) Warsaw (D) Dhaka
- (l) In a research paper, the Methods section should be written in the
(A) future tense (C) present continuous
(B) past tense (D) imperative

- (m) Which is a parameter for judging a journal?
- | | |
|---|-----------------------|
| (A) Page colour | (C) Number of authors |
| ☒ (B) Impact factor | (D) File size |
- (n) Which is the correct capitalisation?
- | | |
|--|---------------------------|
| (A) by theorem 2 | (C) the following Theorem |
| ☒ (B) by Theorem 2 | (D) in figure 3 |
- (o) Which package lets you use `\cref` ?
- | | |
|---|---------------------------|
| (A) <code>hyperref</code> | (C) <code>natbib</code> |
| ☒ (B) <code>cleveref</code> | (D) <code>varioref</code> |

Section B: True or False

(10×1=10)

Write T or F. Correct each false statement in one line.

- (a) A blank line in the source starts a new paragraph. **T**
- (b) `\newcommand` can redefine an existing command such as `\section`. **F** → must use `\renewcommand`
- (c) `\cite` works only if the entry is in the bibliography. **F**
- (d) In TikZ, lines drawn to a named node stop at the node's border. **T**
- (e) The `article` class has a `\chapter` command. **F** → Book/Report has
- (f) `\begin{frame}[allowframebreaks]` lets long content continue on extra slides. **T**
- (g) A corollary usually requires a long independent proof. **F** → follows immediately from a lemma/proof
- (h) You should start a sentence with a symbol when it makes the sentence shorter. **F**
- (i) A good abstract usually contains citations and tables. **F**
- (j) In Pass 2 of reading, you should skip detailed proofs. **T**

Section C: Fill in the Blanks

(10×1=10)

- (a) To make an unnumbered section, write `\section*{4}`.
- (b) The widest-label argument `{9}` is suitable for at most 9 bibliography items.
- (c) The command `\stepcounter` increments a counter and makes it referable by `\label`.
- (d) `\hphantom{x}` keeps only the width of `x`.
- (e) In TikZ, the default unit of coordinates is 1 cm.
- (f) In TikZ, the calc library provides `$(A)!0.5!(B)$`.
- (g) The beamer class option handout collapses overlays for printing.
- (h) According to Peyton Jones, a paper should have exactly one clear idea.
- (i) Author order should be decided early in the writing process.
- (j) A literature search should have high sensitivity and low specificity.

Section D: Matching

(8×1=8)

	Column I		Column II
1	<code>\cmidrule</code>	→ a	repeats a labelled frame
2	<code>\againframe</code>	→ b	partial horizontal rule (booktabs)
3	<code>\pageref</code>	→ c	box of fixed width with wrapped text
4	<code>\parbox</code>	→ d	page number of a label
5	<code>\clip</code>	→ e	restricts later drawing to a region
6	<code>child[missing]</code>	→ f	keeps an empty slot in a tree
7	<code>@inproceedings</code>	→ g	conference paper entry
8	<code>\DeclareMathOperator</code>	→ h	defines an upright operator like argmax

Section E: Predict the Output

(4×3=12)

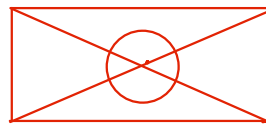
(a) `\begin{enumerate}[label=(\roman*)]`
`\item Alpha`
`\item Beta`
`\item Gamma`
`\end{enumerate}`

(i) Alpha
(ii) Beta
(iii) Gamma

(b) `\begin{tabular}{|c|c|}`
`\hline`
`\multicolumn{2}{|c|}{Total} \\ \hline`
`3 & 4 \\ \hline`
`\end{tabular}`

Total	
3	4

(c) `\begin{tikzpicture}`
`\draw (0,0) rectangle (2,2);`
`\draw (0,0) -- (2,2) (0,2) -- (2,0);`
`\fill (1,1) circle (2pt);`
`\end{tikzpicture}`



(d) `\begin{frame}{Test}`
`Line one \pause`
`Line two \pause`
`\only<2>{Line three}`
`\end{frame}`

Slide-1:
Line One

Slide-2:
Line One
Line Two
Line Three

Slide-3:
Line One
Line Two

How many slides, and what is on each?

Section F: Find and Fix the Errors

(3×4=12)

(a) `\documentclass{article}`
`\usepackage{amsmath}`
`\section{Introduction}`
`\begin{document}`
`The area is $\pi r^2$`
`\end{document}`

→ Inside Document

(b) `\begin{figure}[htbp]`
`\includegraphics[width=5cm]{graph}`
`\caption{A graph}`
`\end{figure}`
`See Figure\ref{fig:graph}.`

→ usepackage {graphics}

→ no label

(c) `\begin{tikzpicture}`
`\draw[->] (0,0) -- (2,0) node[right]{x};`
`\draw (0,0) circle (1);`
`\end{tikzpicture}`

Section G: Correct the Writing

(5×2=10)

- (a) It can be shown that the algorithm terminates.
- (b) n is the number of vertices and we assume $n > 3$.
- (c) All planar graphs have a vertex of degree at most five.
- (d) We use Lemma. 3 to prove the theorem below :
- (e) Our results are really very good and significantly better.

Section H: Short Questions

(6×2=12)

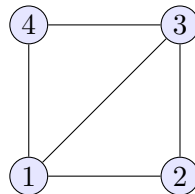
- (a) Explain the difference between `\ref` and `\eqref`. *→ ref without parantheses*
- (b) Why is $$$$...$$$$ discouraged in LaTeX? *→ ref with parantheses*
- (c) What is the purpose of the `every node/.style` option in TikZ? *→ customizing node*
- (d) Give two ways to make beamer slides readable from the back of a room. *① large font*
- (e) What is salami science, and when is splitting a study justified? *② less content*
- (f) What is the difference between a journal and a conference paper? *→ for long manuscript*

papers is split into small papers

Section I: Write the Code

(3×4=12)

- (a) Write a table (with `booktabs`) of three algorithms and their time complexities, with a caption above and a label.
- (b) Draw this graph in TikZ: four nodes on the corners of a 2 cm square, a cycle through them, and one diagonal.



```
\begin{tikzpicture} [v/.style={circle,draw,fill=blue!10}]
\node [v] (a) at (0,0) {1};
\node [v] (b) at (2,0) {2};
\node [v] (c) at (2,2) {3};
\node [v] (d) at (0,2) {4};
\draw (a)--(b)--(c)--(d)--(a) (a)--(c);
\end{tikzpicture}
```

- (c) Define a theorem environment numbered within sections, a lemma sharing its counter, and write one lemma with a proof.

Section J: Long Answer

(1×11=11)

Explain Simon Peyton Jones's seven suggestions for writing a good CS paper. For "Nail your contributions to the mast", give one vague and one refutable contribution statement.

I. @ \begin{table} [htbp]

\centering

\caption{Time complexity of sorting algorithm} \label{tab:sort}

\begin{tabular}{@{}ll@{}}

\toprule

Algorithm & Time

\midrule

Merge Sort & $O(n \log n)$

Bubble Sort & $O(n^2)$

Quick Sort & $O(n^2)$

\bottomrule

\end{tabular}

\end{table}